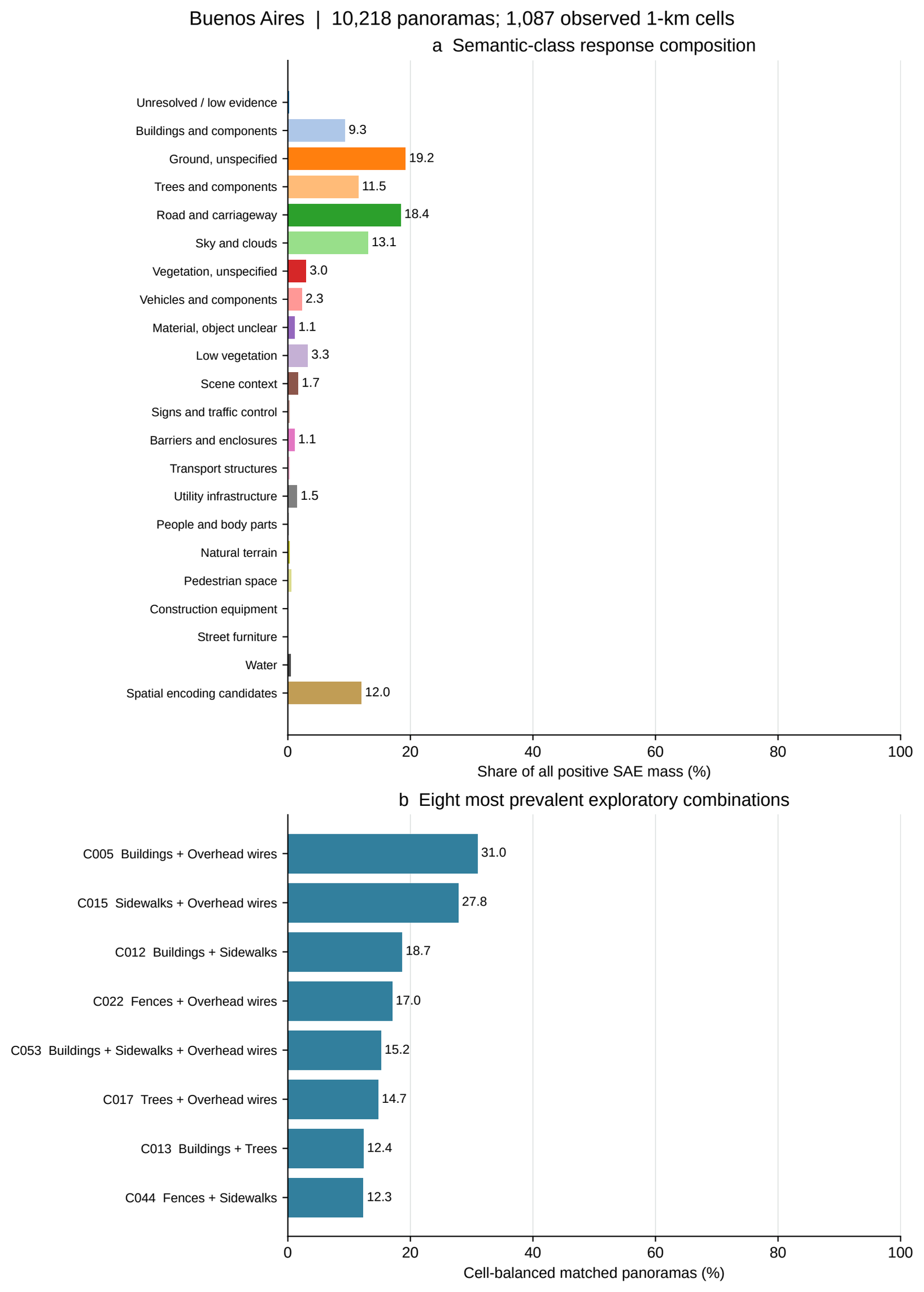
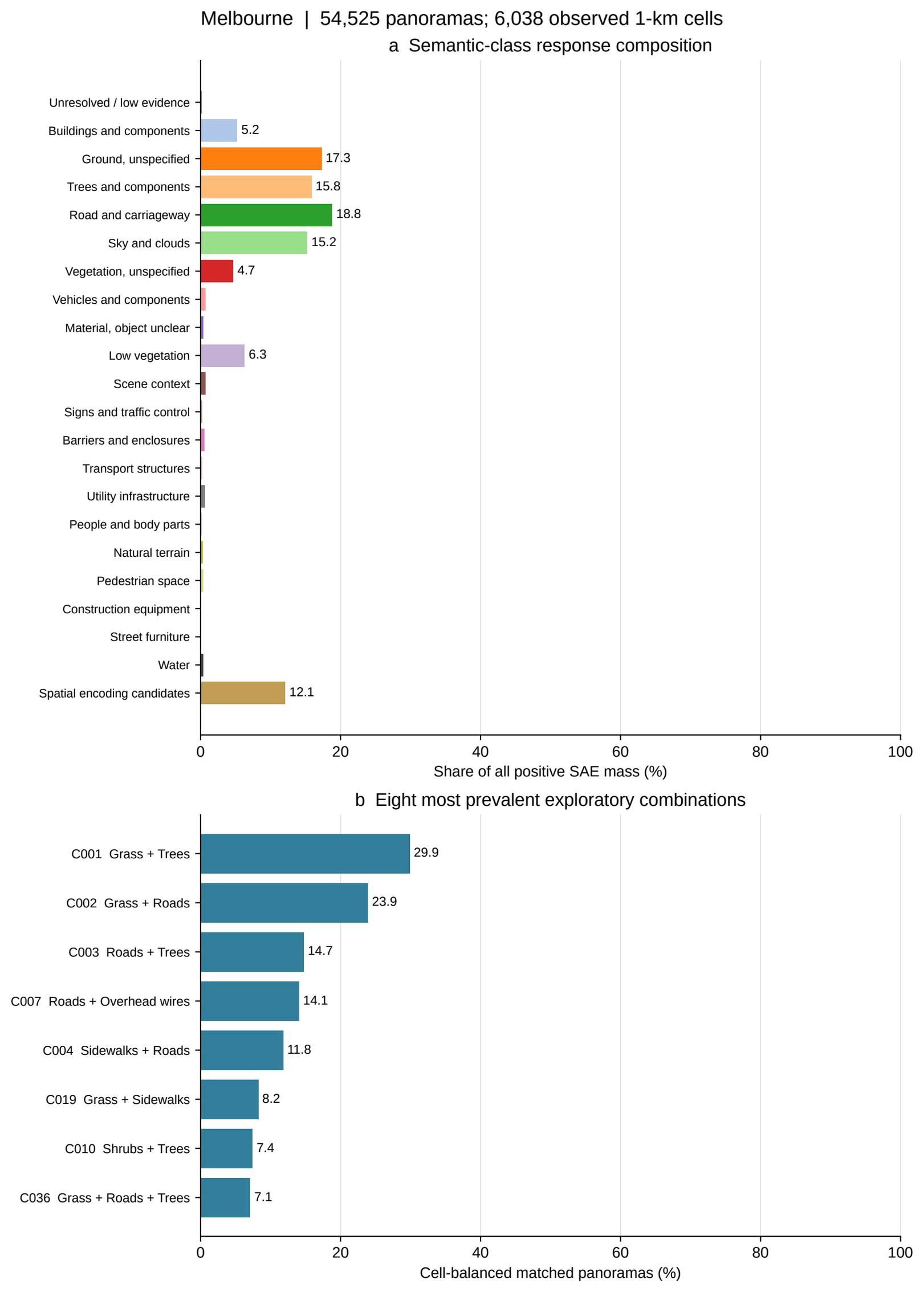
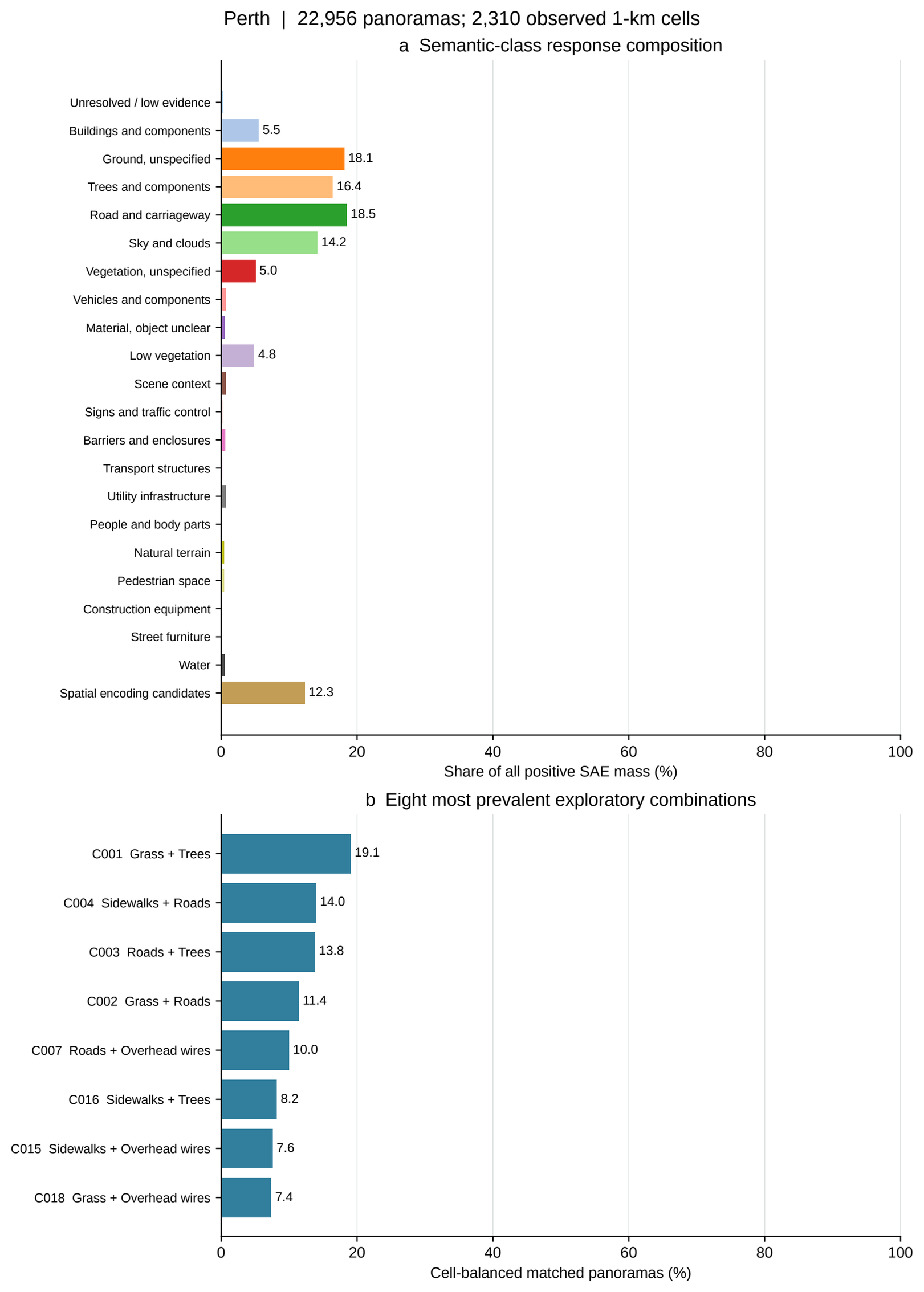
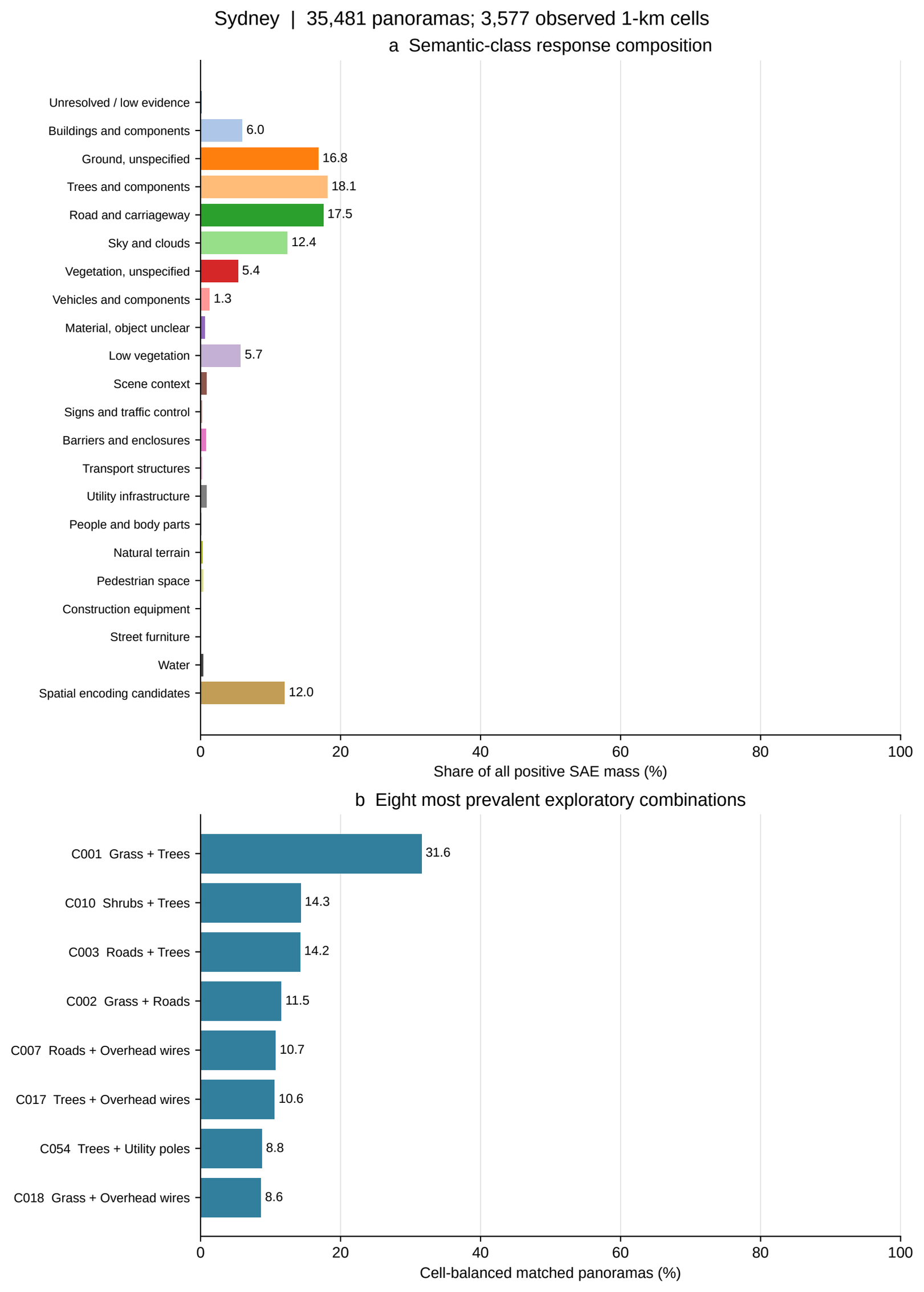
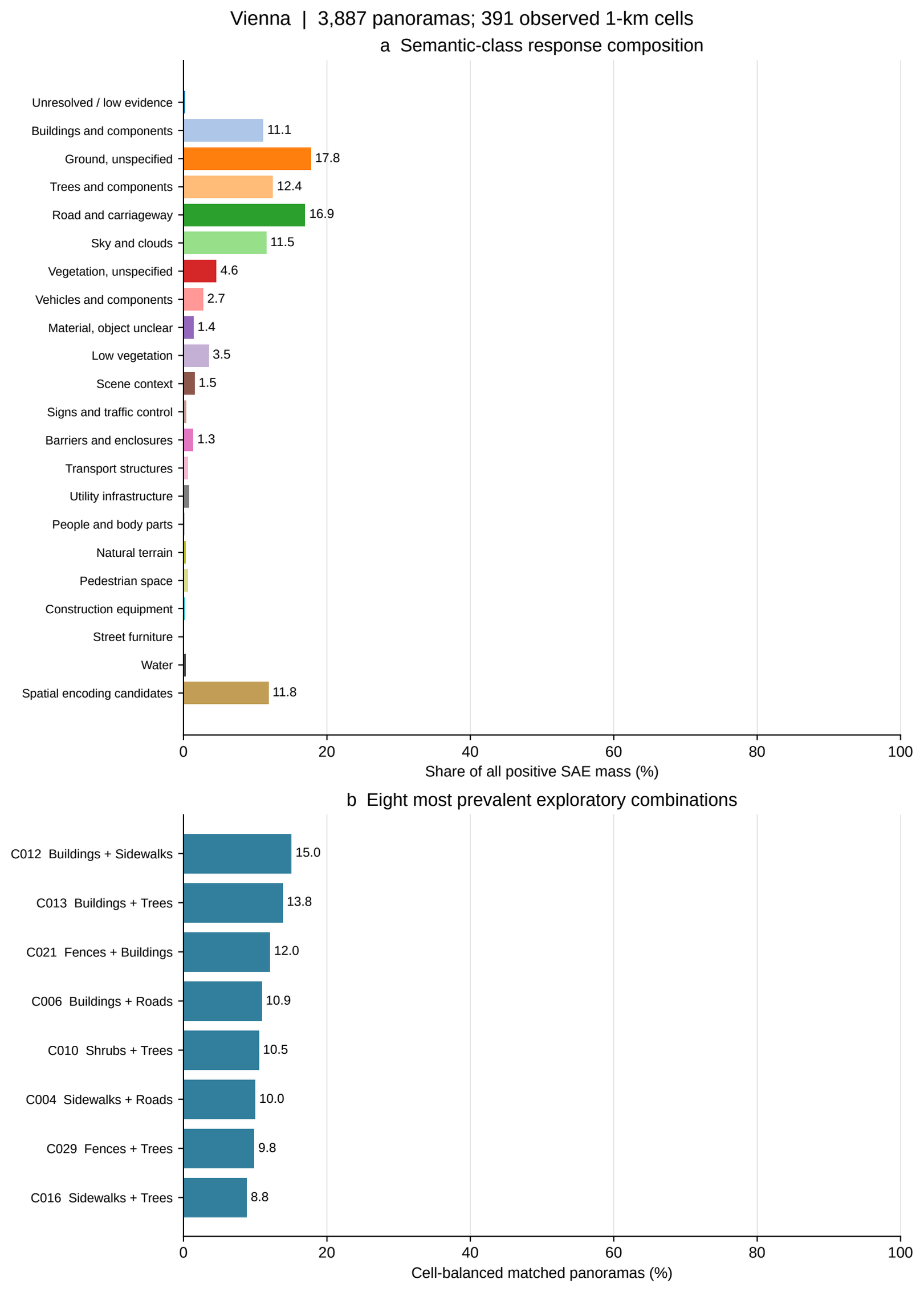
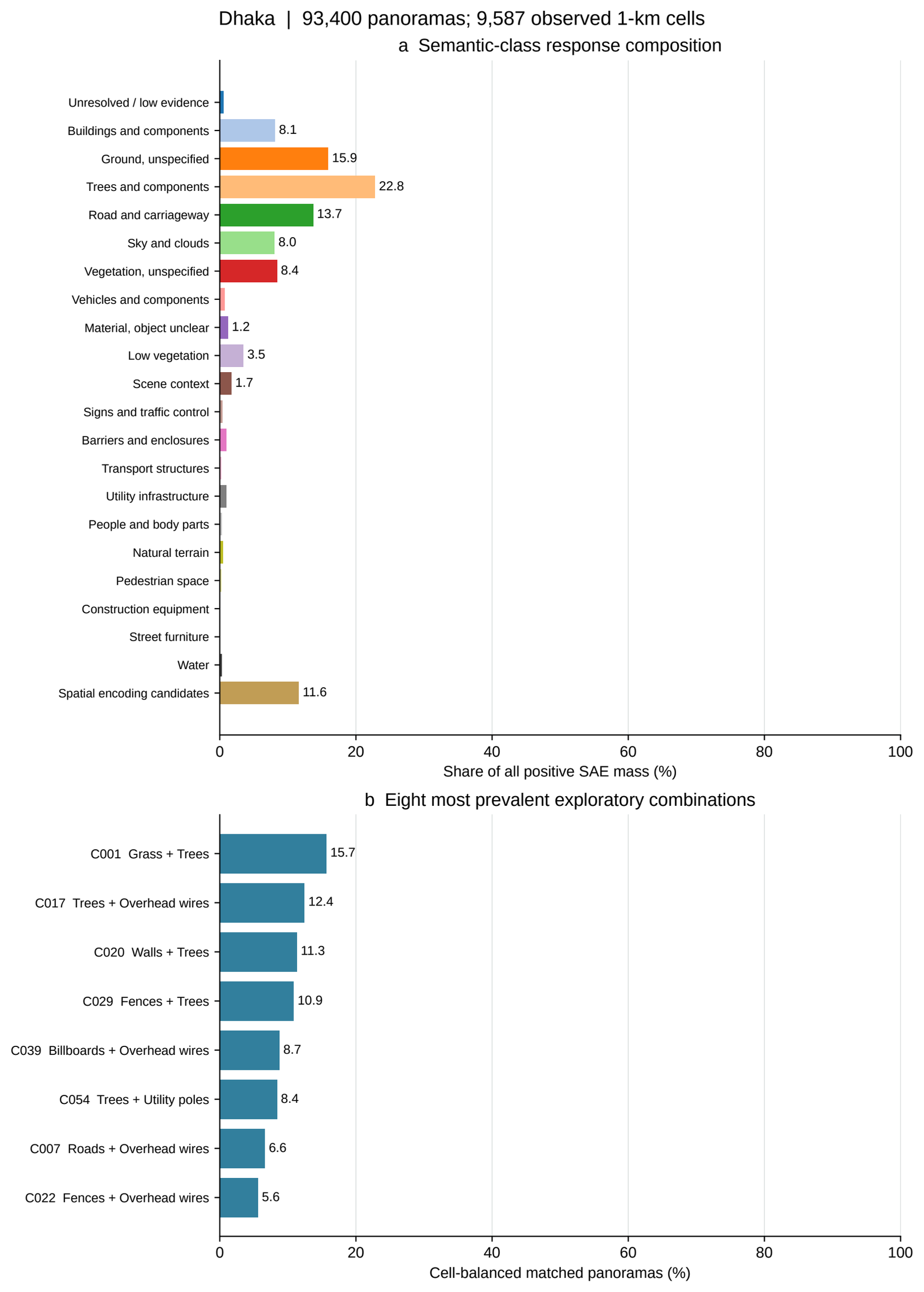

Buenos Aires | 10,218 panoramas; 1,087 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
9.3
Ground, unspecified
19.2
Trees and components
11.5
Road and carriageway
18.4
Sky and clouds
13.1
Vegetation, unspecified
3.0
Vehicles and components
2.3
Material, object unclear
1.1
Low vegetation
3.3
Scene context
1.7
Signs and traffic control
Barriers and enclosures
1.1
Transport structures
Utility infrastructure
1.5
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C005 Buildings + Overhead wires
31.0
C015 Sidewalks + Overhead wires
27.8
C012 Buildings + Sidewalks
18.7
C022 Fences + Overhead wires
17.0
C053 Buildings + Sidewalks + Overhead wires
15.2
C017 Trees + Overhead wires
14.7
C013 Buildings + Trees
12.4
C044 Fences + Sidewalks
12.3
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

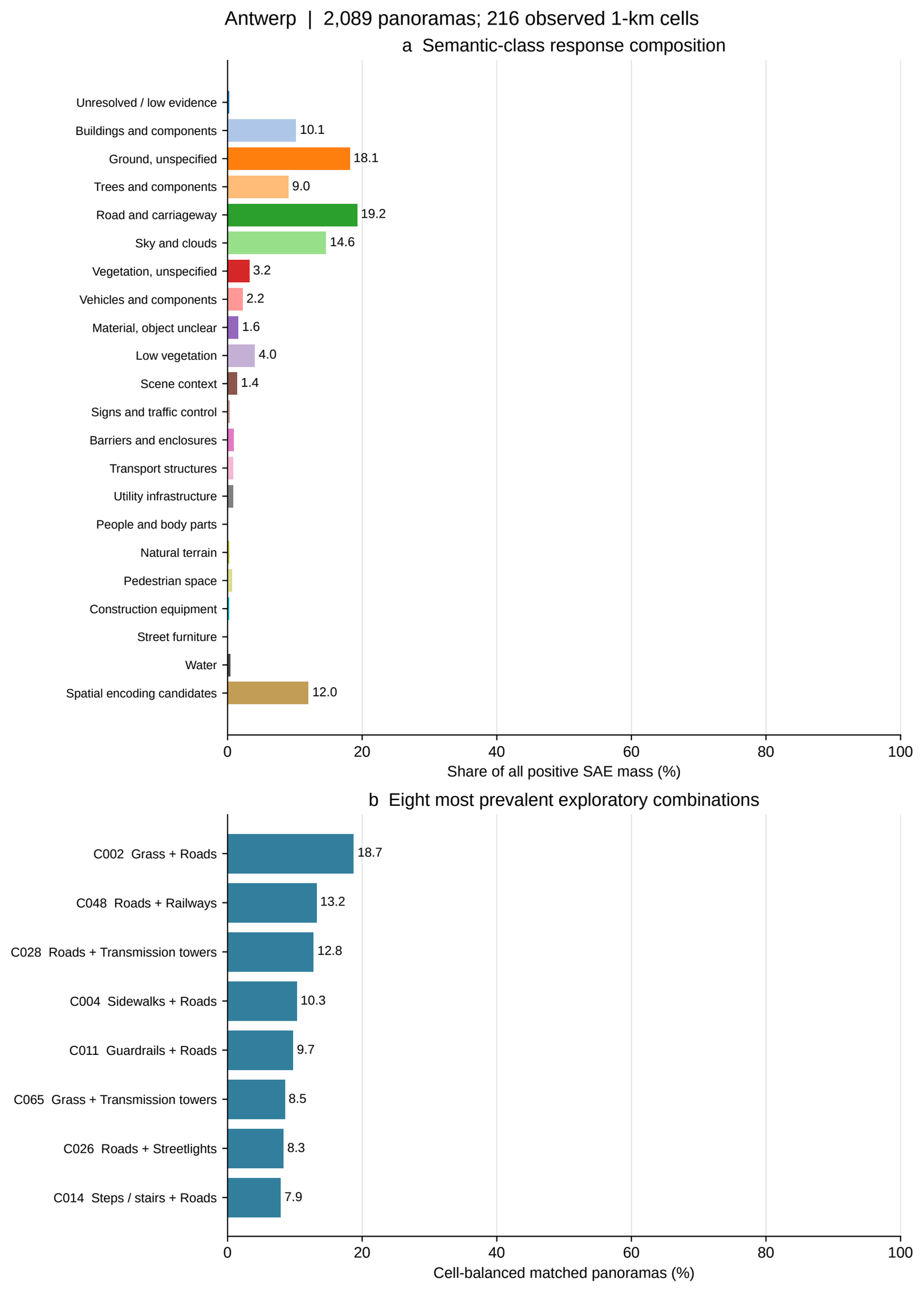

Melbourne | 54,525 panoramas; 6,038 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
5.2
Ground, unspecified
17.3
Trees and components
15.8
Road and carriageway
18.8
Sky and clouds
15.2
Vegetation, unspecified
4.7
Vehicles and components
Material, object unclear
Low vegetation
6.3
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.1
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
29.9
C002 Grass + Roads
23.9
C003 Roads + Trees
14.7
C007 Roads + Overhead wires
14.1
C004 Sidewalks + Roads
11.8
C019 Grass + Sidewalks
8.2
C010 Shrubs + Trees
7.4
C036 Grass + Roads + Trees
7.1
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

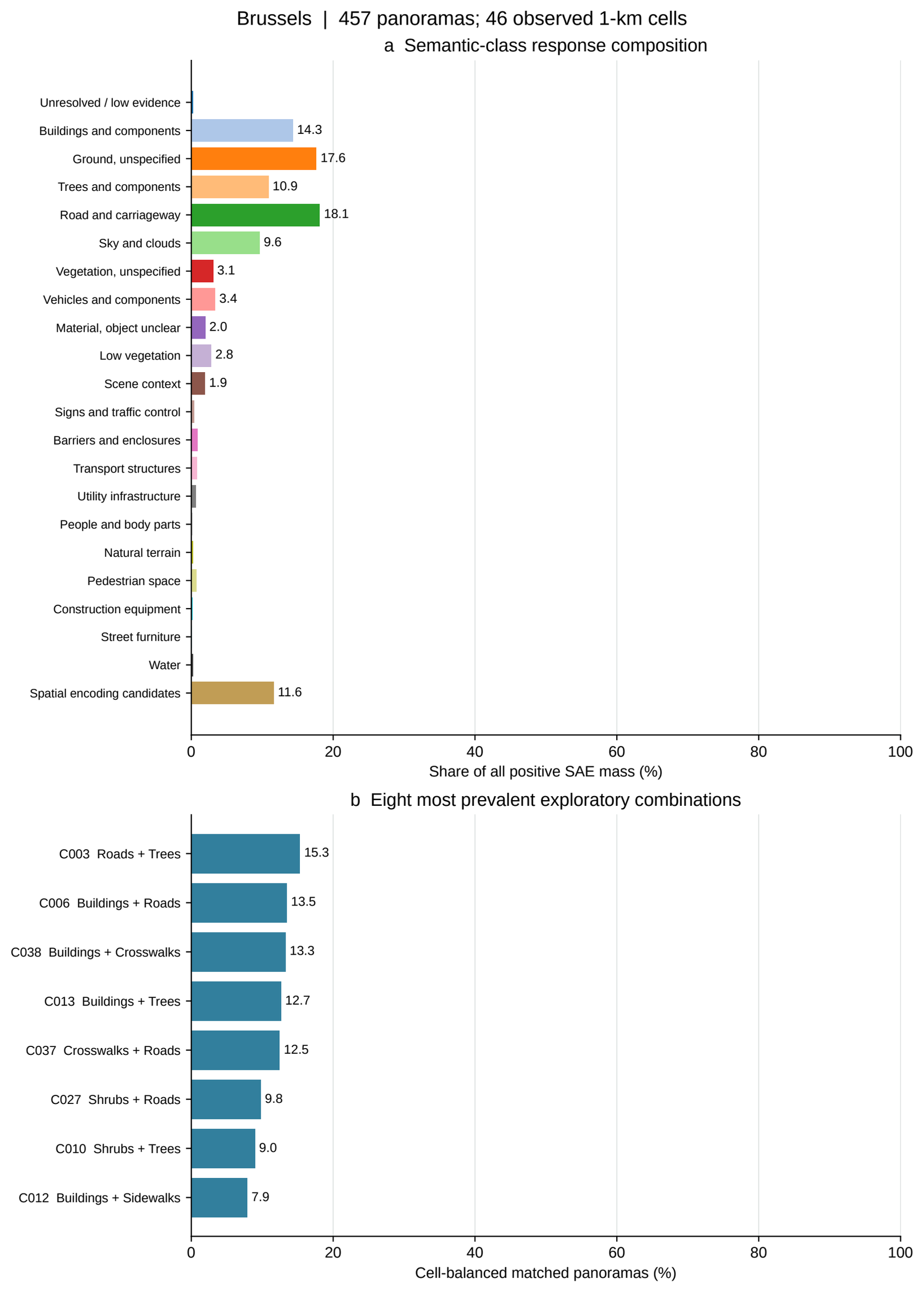

Perth | 22,956 panoramas; 2,310 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
5.5
Ground, unspecified
18.1
Trees and components
16.4
Road and carriageway
18.5
Sky and clouds
14.2
Vegetation, unspecified
5.0
Vehicles and components
Material, object unclear
Low vegetation
4.8
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.3
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
19.1
C004 Sidewalks + Roads
14.0
C003 Roads + Trees
13.8
C002 Grass + Roads
11.4
C007 Roads + Overhead wires
10.0
C016 Sidewalks + Trees
8.2
C015 Sidewalks + Overhead wires
7.6
C018 Grass + Overhead wires
7.4
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

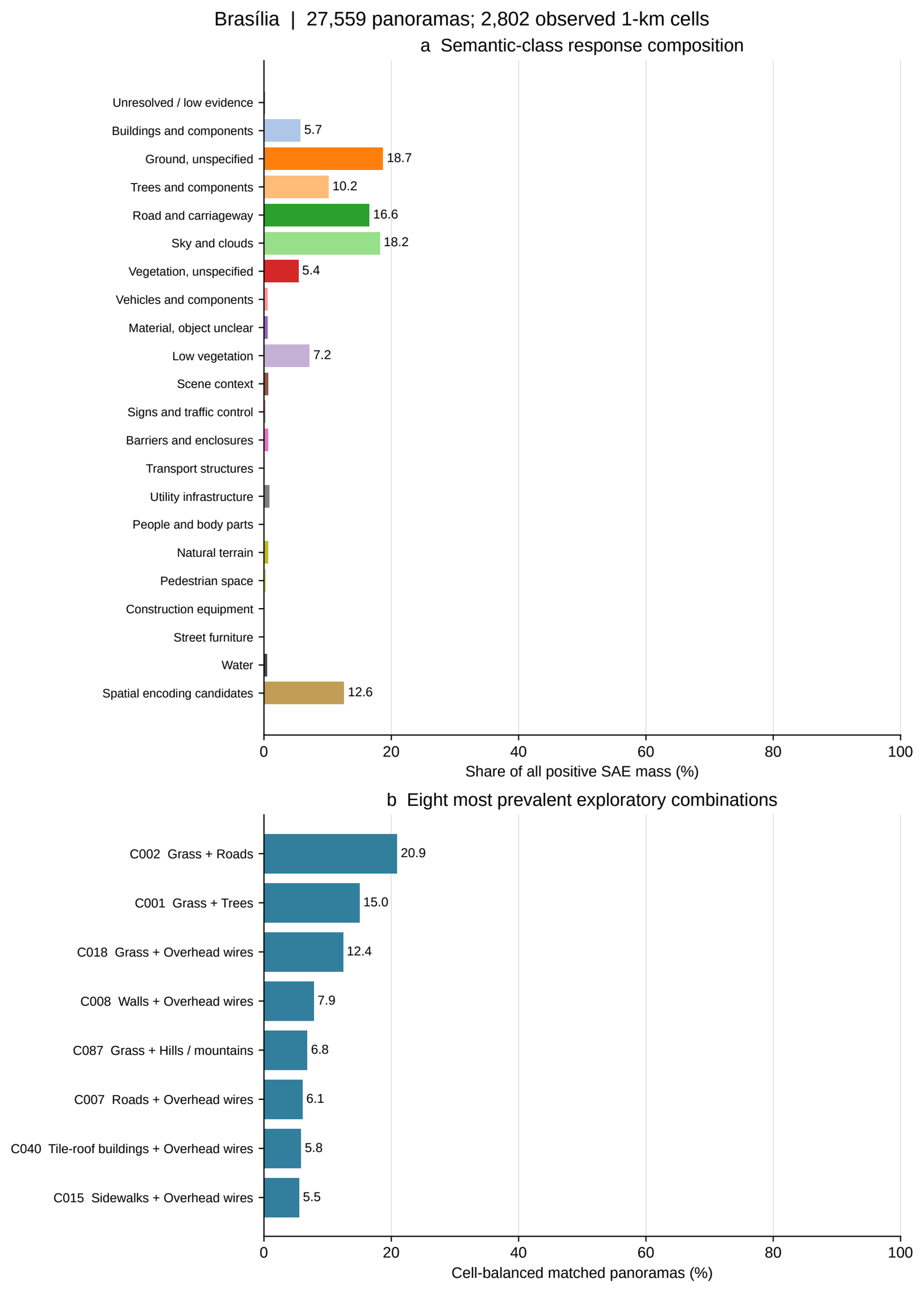

Sydney | 35,481 panoramas; 3,577 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.0
Ground, unspecified
16.8
Trees and components
18.1
Road and carriageway
17.5
Sky and clouds
12.4
Vegetation, unspecified
5.4
Vehicles and components
1.3
Material, object unclear
Low vegetation
5.7
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
31.6
C010 Shrubs + Trees
14.3
C003 Roads + Trees
14.2
C002 Grass + Roads
11.5
C007 Roads + Overhead wires
10.7
C017 Trees + Overhead wires
10.6
C054 Trees + Utility poles
8.8
C018 Grass + Overhead wires
8.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

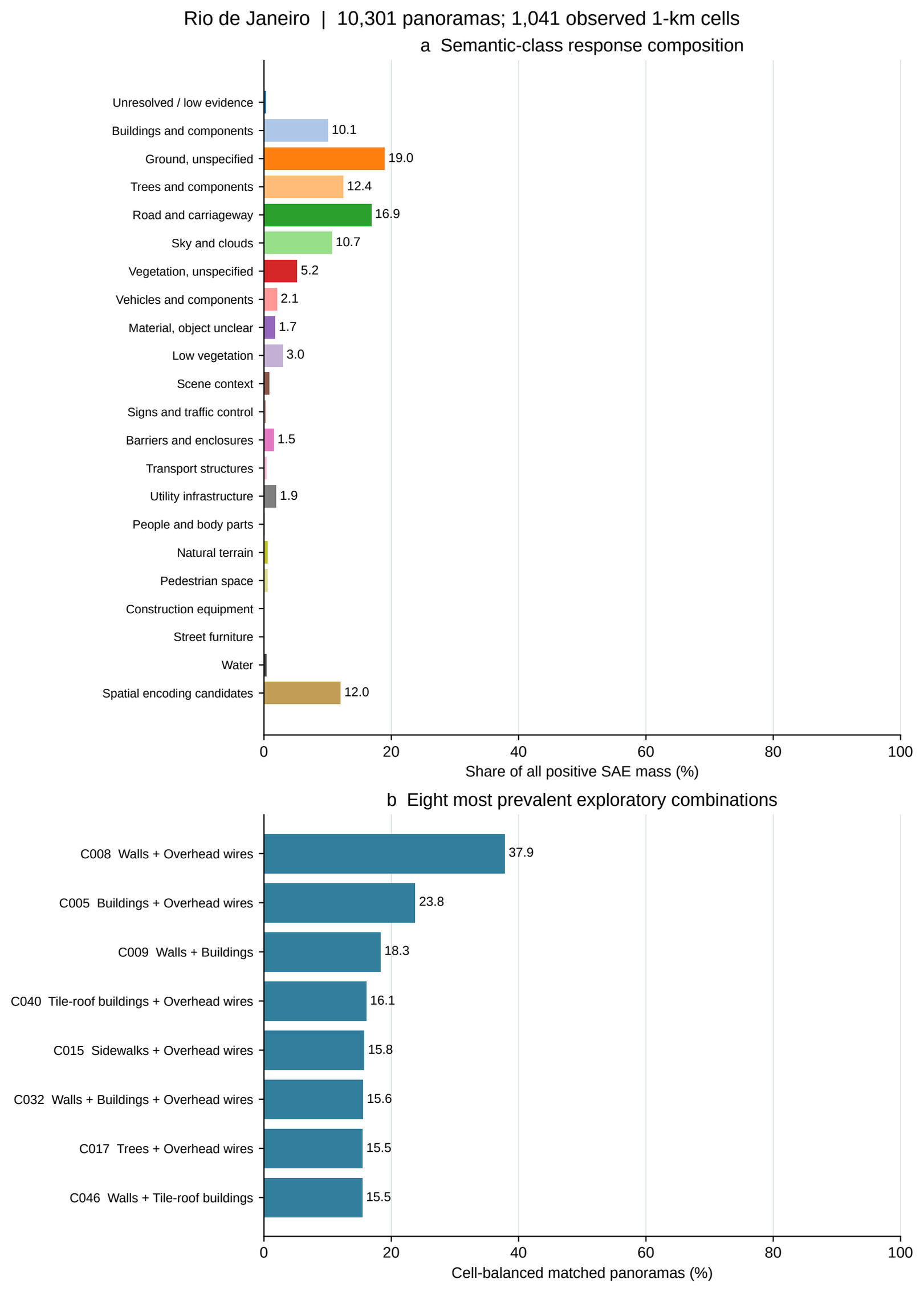

Vienna | 3,887 panoramas; 391 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
11.1
Ground, unspecified
17.8
Trees and components
12.4
Road and carriageway
16.9
Sky and clouds
11.5
Vegetation, unspecified
4.6
Vehicles and components
2.7
Material, object unclear
1.4
Low vegetation
3.5
Scene context
1.5
Signs and traffic control
Barriers and enclosures
1.3
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.8
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C012 Buildings + Sidewalks
15.0
C013 Buildings + Trees
13.8
C021 Fences + Buildings
12.0
C006 Buildings + Roads
10.9
C010 Shrubs + Trees
10.5
C004 Sidewalks + Roads
10.0
C029 Fences + Trees
9.8
C016 Sidewalks + Trees
8.8
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

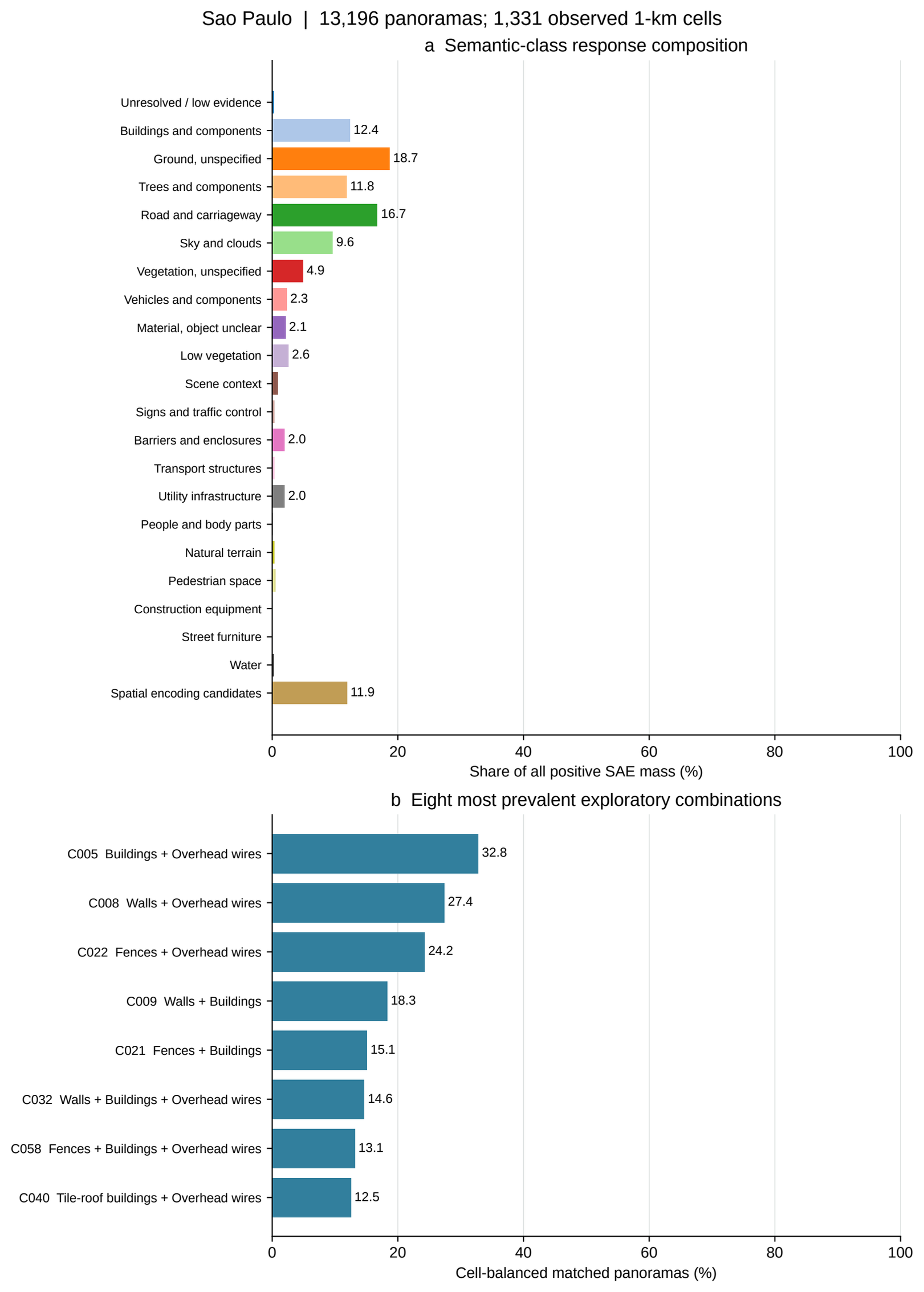

Dhaka | 93,400 panoramas; 9,587 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
8.1
Ground, unspecified
15.9
Trees and components
22.8
Road and carriageway
13.7
Sky and clouds
8.0
Vegetation, unspecified
8.4
Vehicles and components
Material, object unclear
1.2
Low vegetation
3.5
Scene context
1.7
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.6
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
15.7
C017 Trees + Overhead wires
12.4
C020 Walls + Trees
11.3
C029 Fences + Trees
10.9
C039 Billboards + Overhead wires
8.7
C054 Trees + Utility poles
8.4
C007 Roads + Overhead wires
6.6
C022 Fences + Overhead wires
5.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

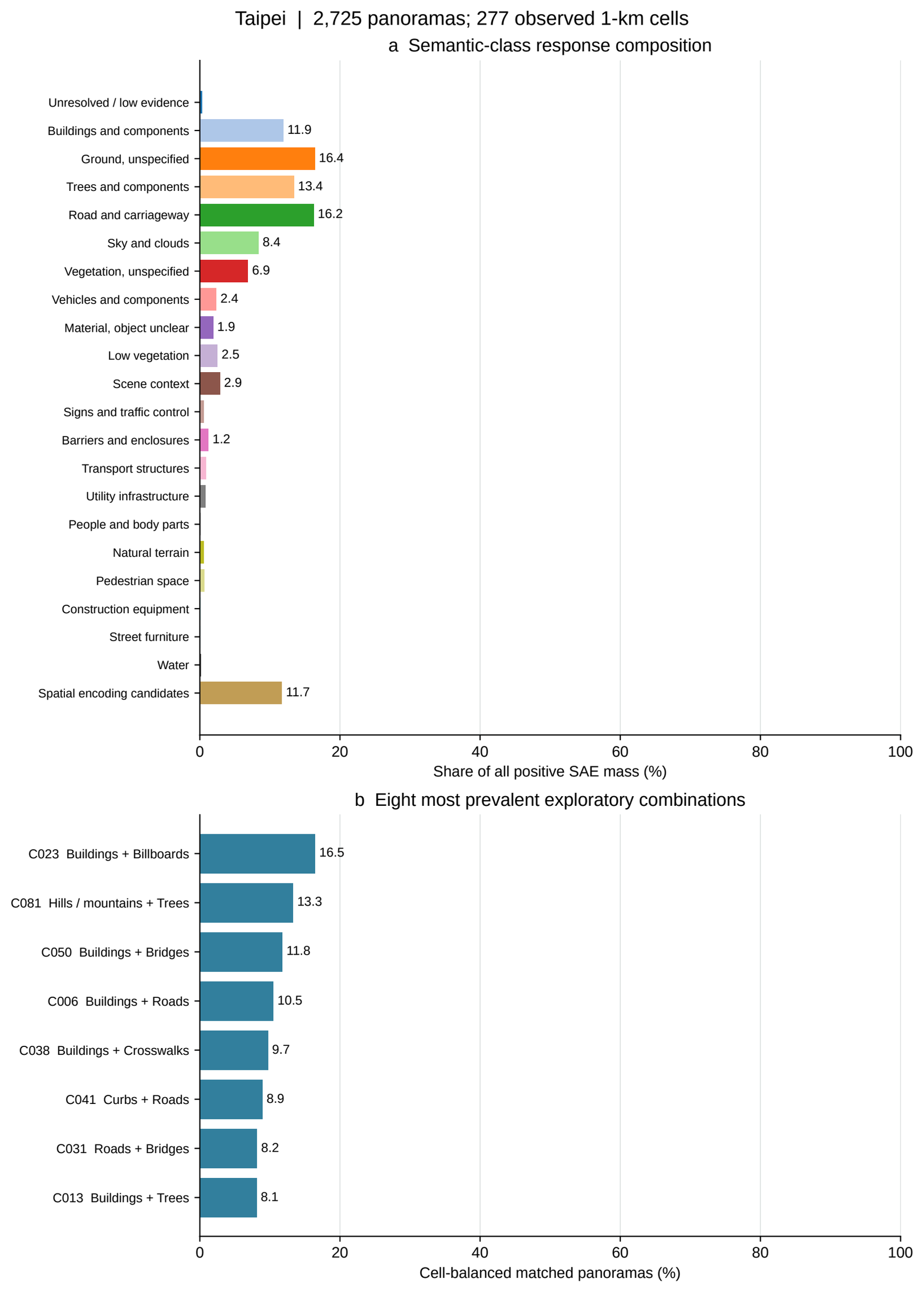

Antwerp | 2,089 panoramas; 216 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
10.1
Ground, unspecified
18.1
Trees and components
9.0
Road and carriageway
19.2
Sky and clouds
14.6
Vegetation, unspecified
3.2
Vehicles and components
2.2
Material, object unclear
1.6
Low vegetation
4.0
Scene context
1.4
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C002 Grass + Roads
18.7
C048 Roads + Railways
13.2
C028 Roads + Transmission towers
12.8
C004 Sidewalks + Roads
10.3
C011 Guardrails + Roads
9.7
C065 Grass + Transmission towers
8.5
C026 Roads + Streetlights
8.3
C014 Steps / stairs + Roads
7.9
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

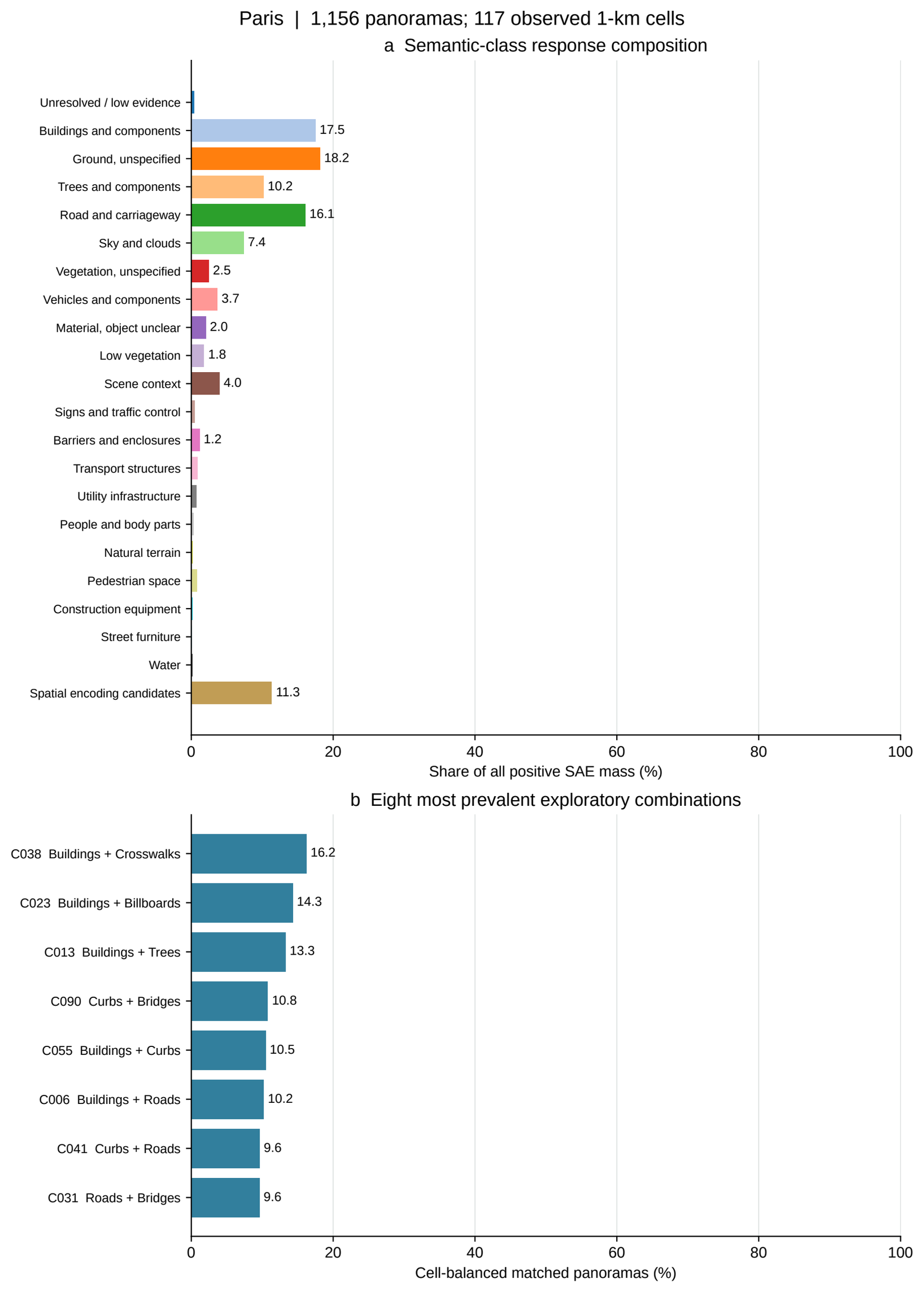

Brussels | 457 panoramas; 46 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
14.3
Ground, unspecified
17.6
Trees and components
10.9
Road and carriageway
18.1
Sky and clouds
9.6
Vegetation, unspecified
3.1
Vehicles and components
3.4
Material, object unclear
2.0
Low vegetation
2.8
Scene context
1.9
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.6
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C003 Roads + Trees
15.3
C006 Buildings + Roads
13.5
C038 Buildings + Crosswalks
13.3
C013 Buildings + Trees
12.7
C037 Crosswalks + Roads
12.5
C027 Shrubs + Roads
9.8
C010 Shrubs + Trees
9.0
C012 Buildings + Sidewalks
7.9
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

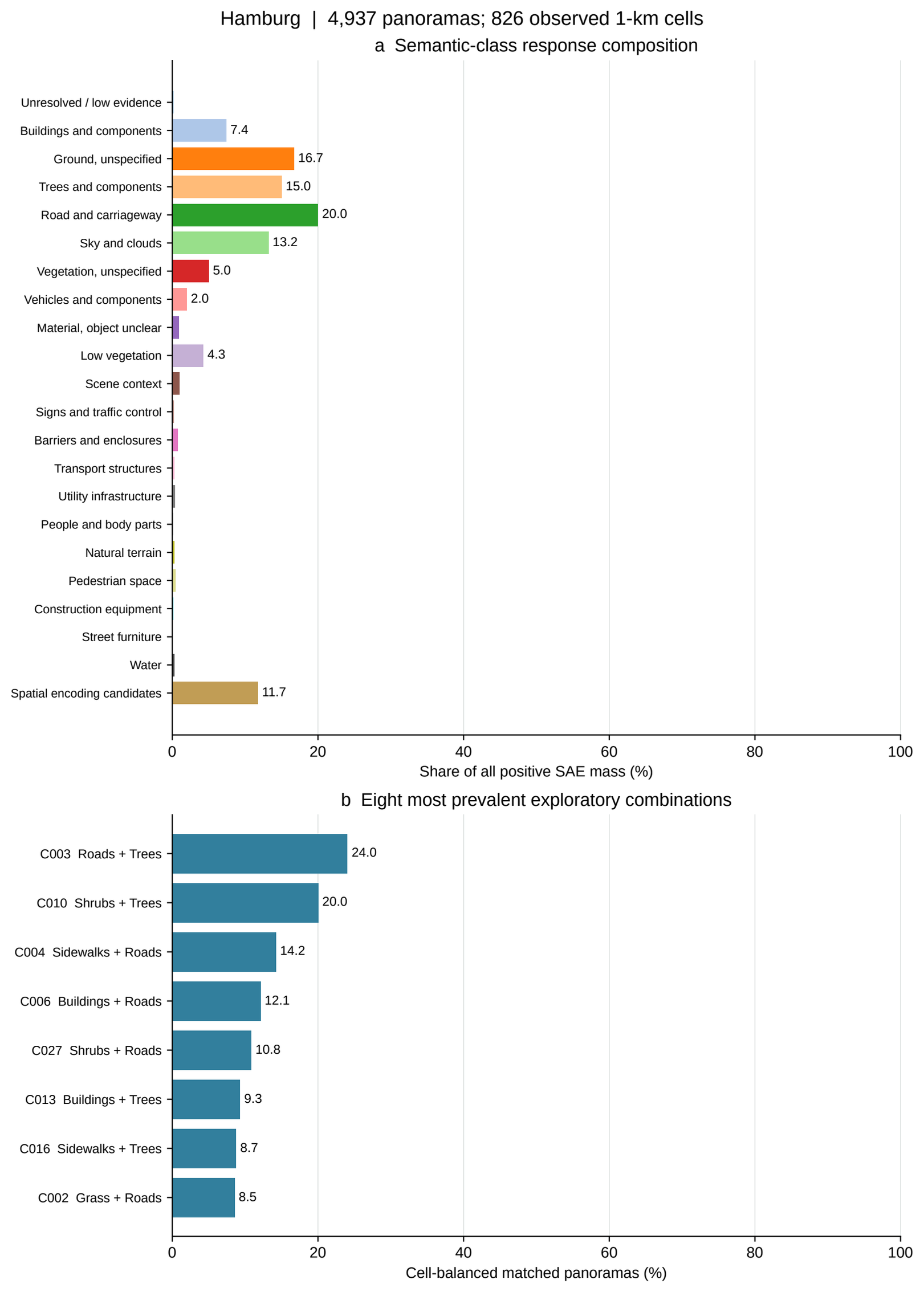

Brasília | 27,559 panoramas; 2,802 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
5.7
Ground, unspecified
18.7
Trees and components
10.2
Road and carriageway
16.6
Sky and clouds
18.2
Vegetation, unspecified
5.4
Vehicles and components
Material, object unclear
Low vegetation
7.2
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.6
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C002 Grass + Roads
20.9
C001 Grass + Trees
15.0
C018 Grass + Overhead wires
12.4
C008 Walls + Overhead wires
7.9
C087 Grass + Hills / mountains
6.8
C007 Roads + Overhead wires
6.1
C040 Tile-roof buildings + Overhead wires
5.8
C015 Sidewalks + Overhead wires
5.5
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

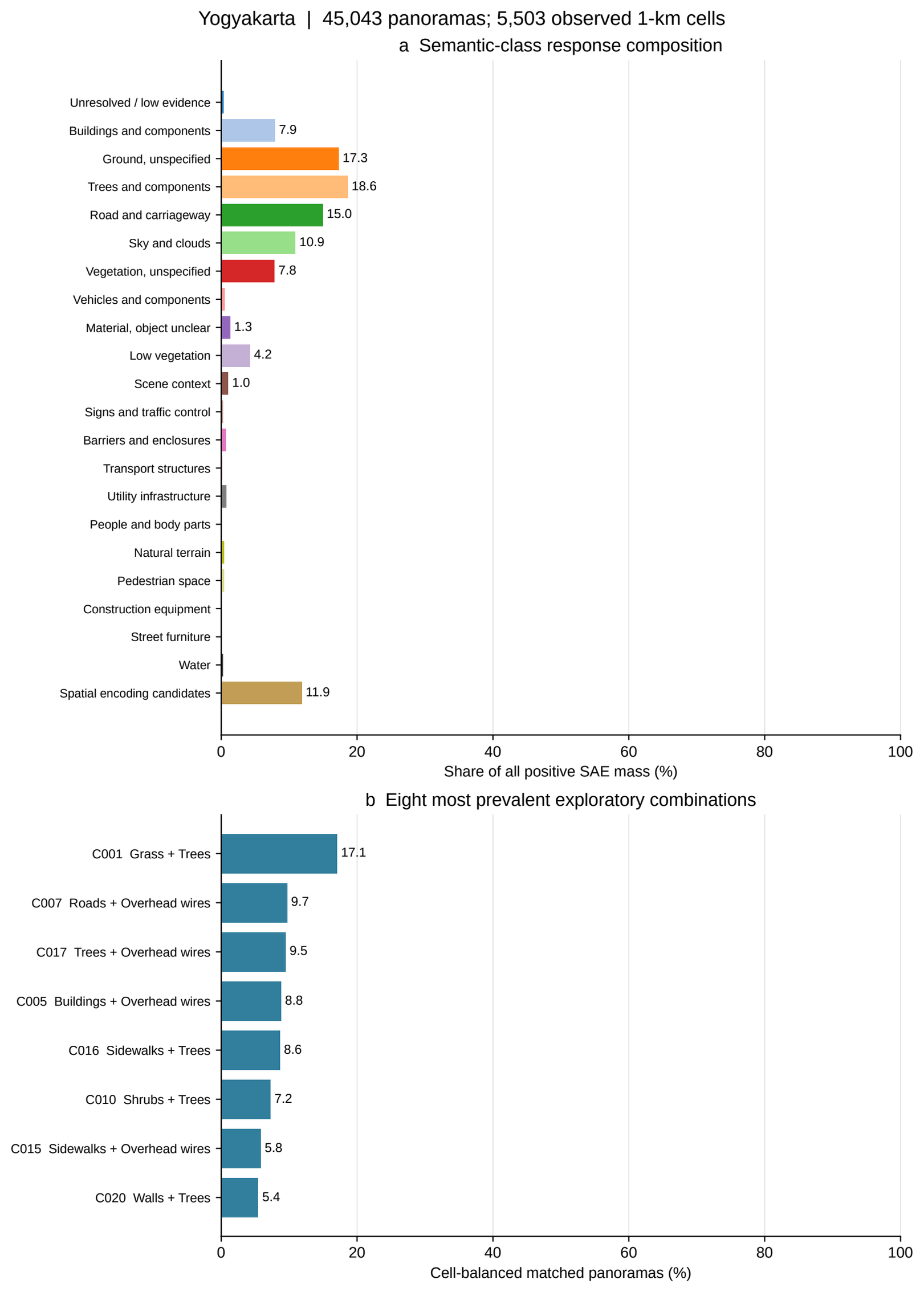

Rio de Janeiro | 10,301 panoramas; 1,041 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
10.1
Ground, unspecified
19.0
Trees and components
12.4
Road and carriageway
16.9
Sky and clouds
10.7
Vegetation, unspecified
5.2
Vehicles and components
2.1
Material, object unclear
1.7
Low vegetation
3.0
Scene context
Signs and traffic control
Barriers and enclosures
1.5
Transport structures
Utility infrastructure
1.9
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C008 Walls + Overhead wires
37.9
C005 Buildings + Overhead wires
23.8
C009 Walls + Buildings
18.3
C040 Tile-roof buildings + Overhead wires
16.1
C015 Sidewalks + Overhead wires
15.8
C032 Walls + Buildings + Overhead wires
15.6
C017 Trees + Overhead wires
15.5
C046 Walls + Tile-roof buildings
15.5
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

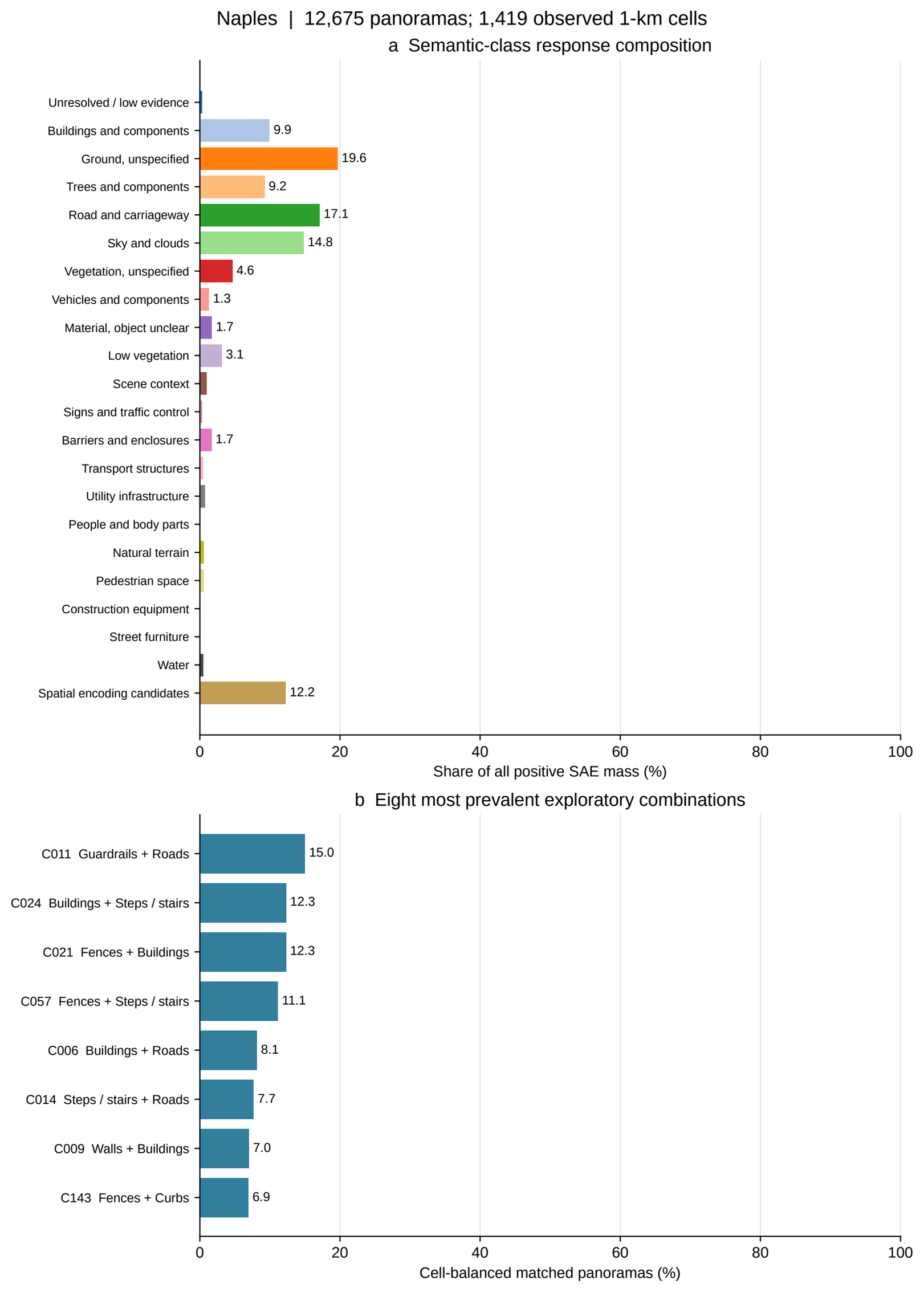

Sao Paulo | 13,196 panoramas; 1,331 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
12.4
Ground, unspecified
18.7
Trees and components
11.8
Road and carriageway
16.7
Sky and clouds
9.6
Vegetation, unspecified
4.9
Vehicles and components
2.3
Material, object unclear
2.1
Low vegetation
2.6
Scene context
Signs and traffic control
Barriers and enclosures
2.0
Transport structures
Utility infrastructure
2.0
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.9
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C005 Buildings + Overhead wires
32.8
C008 Walls + Overhead wires
27.4
C022 Fences + Overhead wires
24.2
C009 Walls + Buildings
18.3
C021 Fences + Buildings
15.1
C032 Walls + Buildings + Overhead wires
14.6
C058 Fences + Buildings + Overhead wires
13.1
C040 Tile-roof buildings + Overhead wires
12.5
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

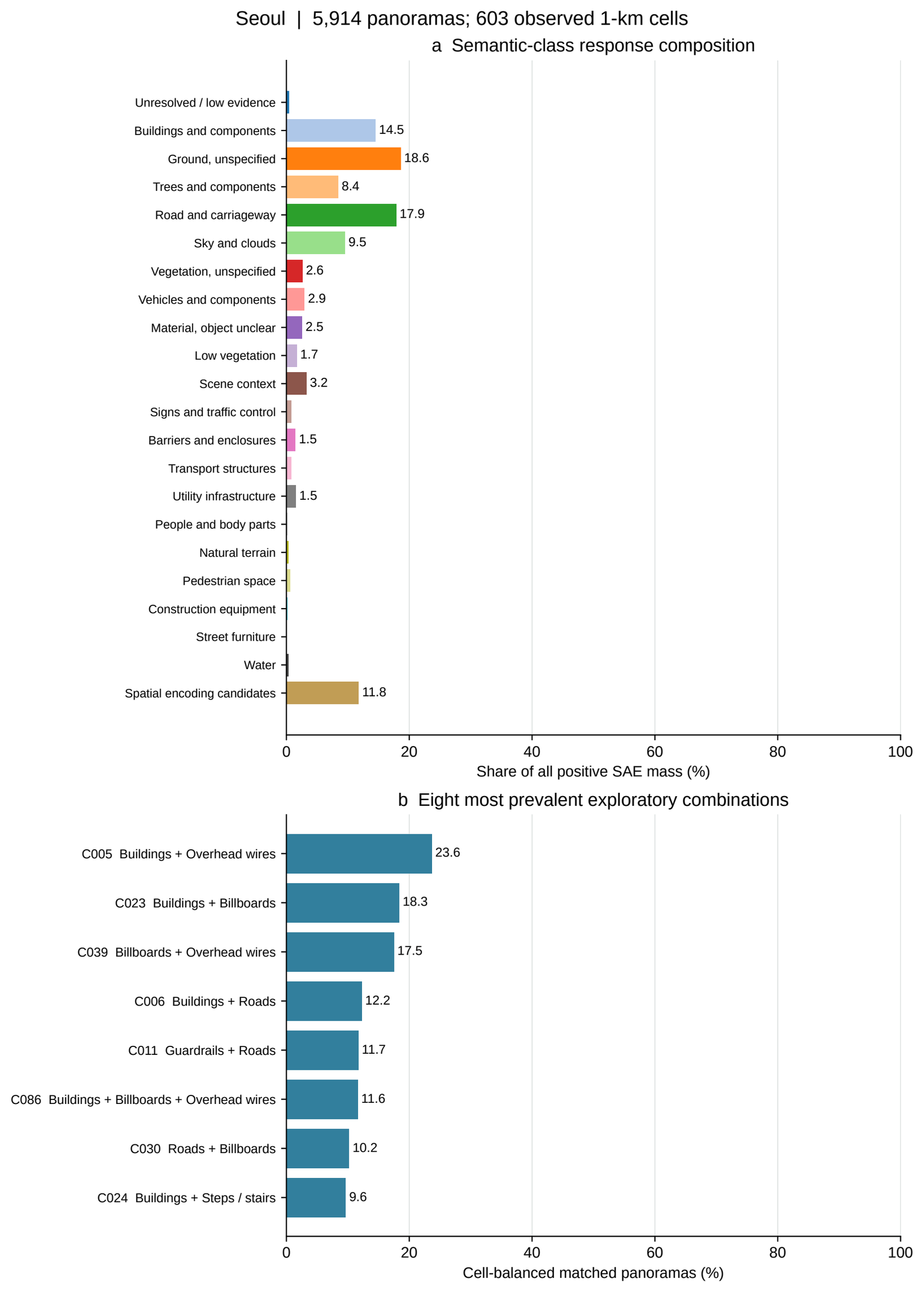

Taipei | 2,725 panoramas; 277 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
11.9
Ground, unspecified
16.4
Trees and components
13.4
Road and carriageway
16.2
Sky and clouds
8.4
Vegetation, unspecified
6.9
Vehicles and components
2.4
Material, object unclear
1.9
Low vegetation
2.5
Scene context
2.9
Signs and traffic control
Barriers and enclosures
1.2
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.7
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C023 Buildings + Billboards
16.5
C081 Hills / mountains + Trees
13.3
C050 Buildings + Bridges
11.8
C006 Buildings + Roads
10.5
C038 Buildings + Crosswalks
9.7
C041 Curbs + Roads
8.9
C031 Roads + Bridges
8.2
C013 Buildings + Trees
8.1
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

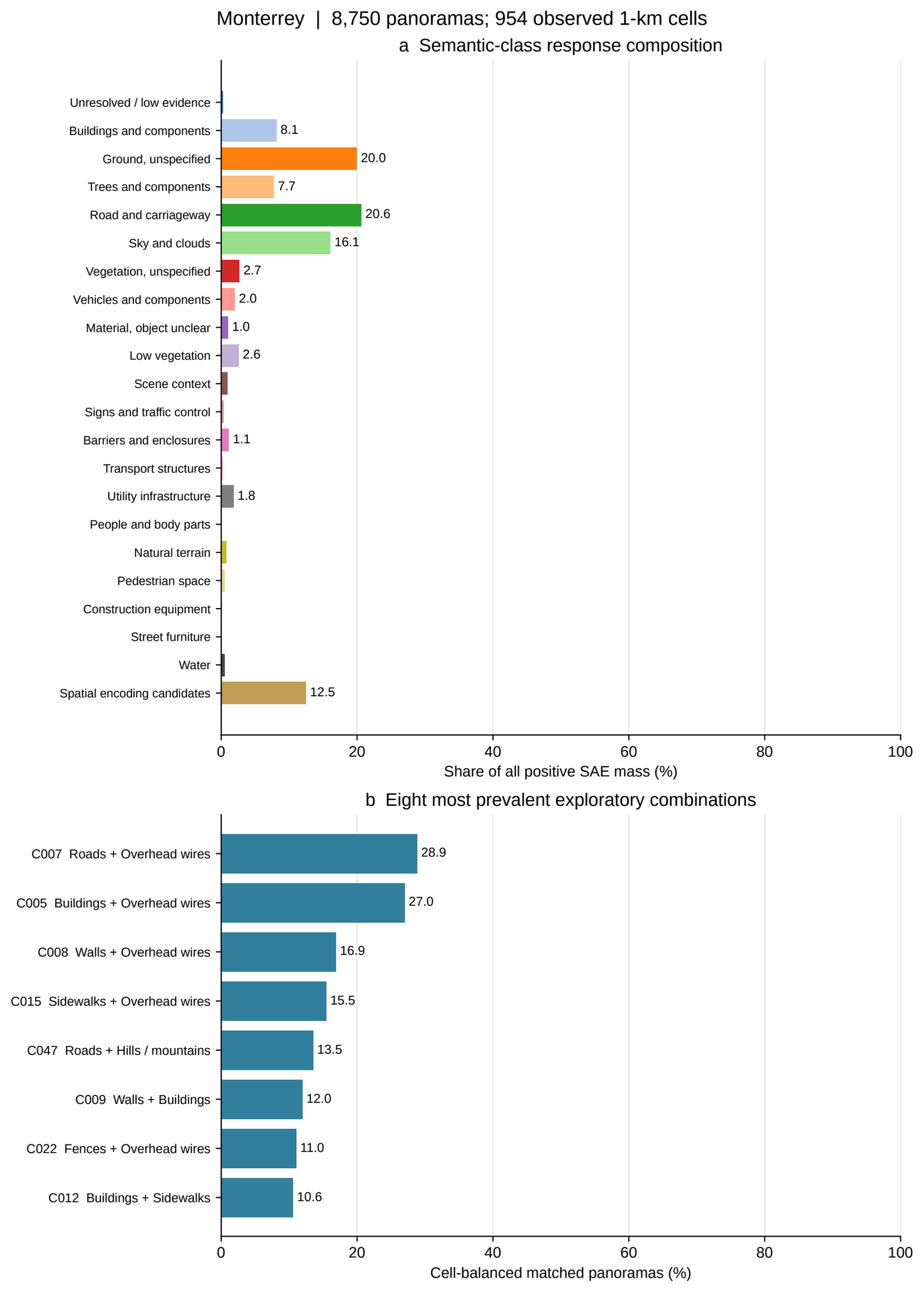

Paris | 1,156 panoramas; 117 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
17.5
Ground, unspecified
18.2
Trees and components
10.2
Road and carriageway
16.1
Sky and clouds
7.4
Vegetation, unspecified
2.5
Vehicles and components
3.7
Material, object unclear
2.0
Low vegetation
1.8
Scene context
4.0
Signs and traffic control
Barriers and enclosures
1.2
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.3
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C038 Buildings + Crosswalks
16.2
C023 Buildings + Billboards
14.3
C013 Buildings + Trees
13.3
C090 Curbs + Bridges
10.8
C055 Buildings + Curbs
10.5
C006 Buildings + Roads
10.2
C041 Curbs + Roads
9.6
C031 Roads + Bridges
9.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

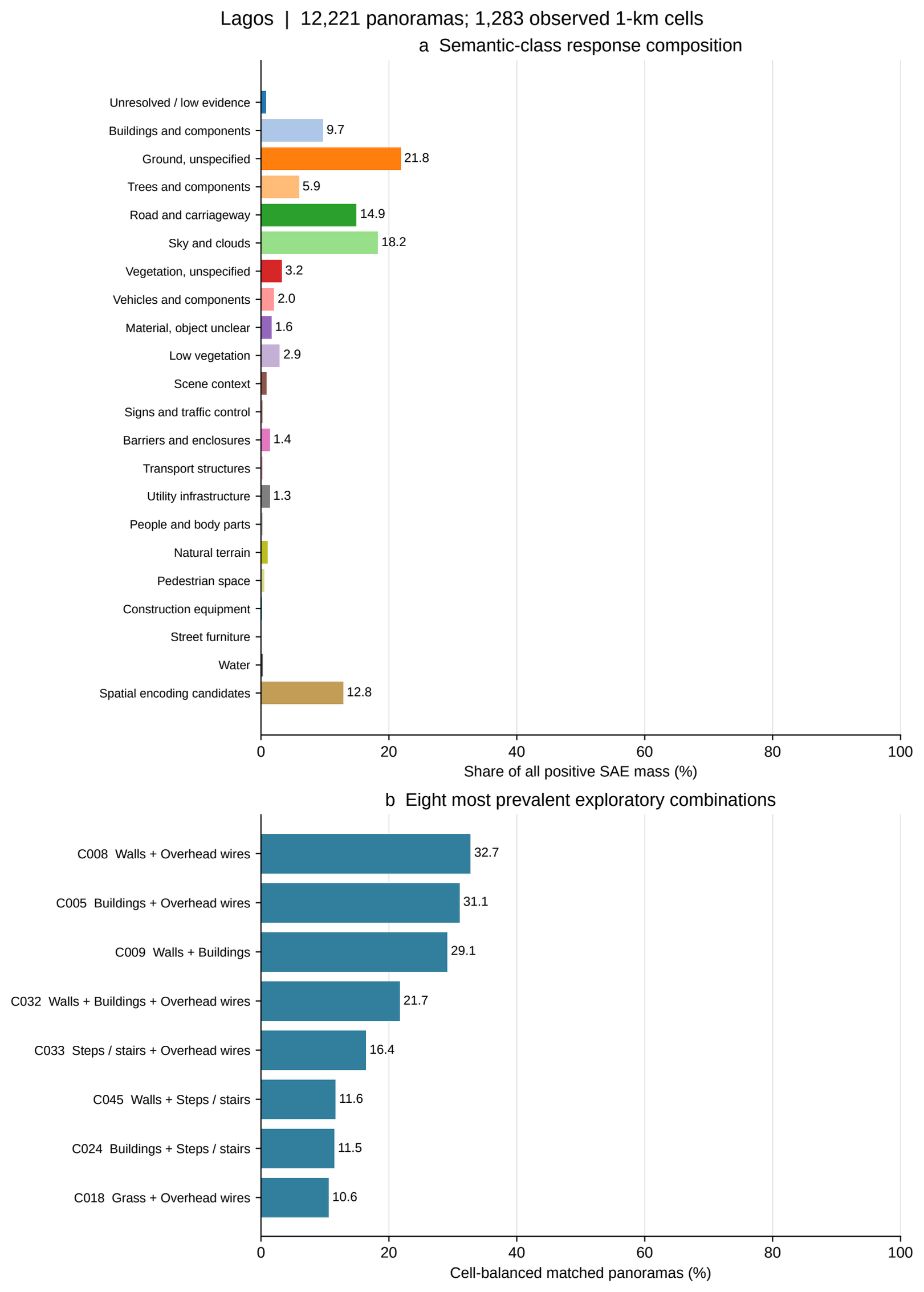

Hamburg | 4,937 panoramas; 826 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
7.4
Ground, unspecified
16.7
Trees and components
15.0
Road and carriageway
20.0
Sky and clouds
13.2
Vegetation, unspecified
5.0
Vehicles and components
2.0
Material, object unclear
Low vegetation
4.3
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.7
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C003 Roads + Trees
24.0
C010 Shrubs + Trees
20.0
C004 Sidewalks + Roads
14.2
C006 Buildings + Roads
12.1
C027 Shrubs + Roads
10.8
C013 Buildings + Trees
9.3
C016 Sidewalks + Trees
8.7
C002 Grass + Roads
8.5
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

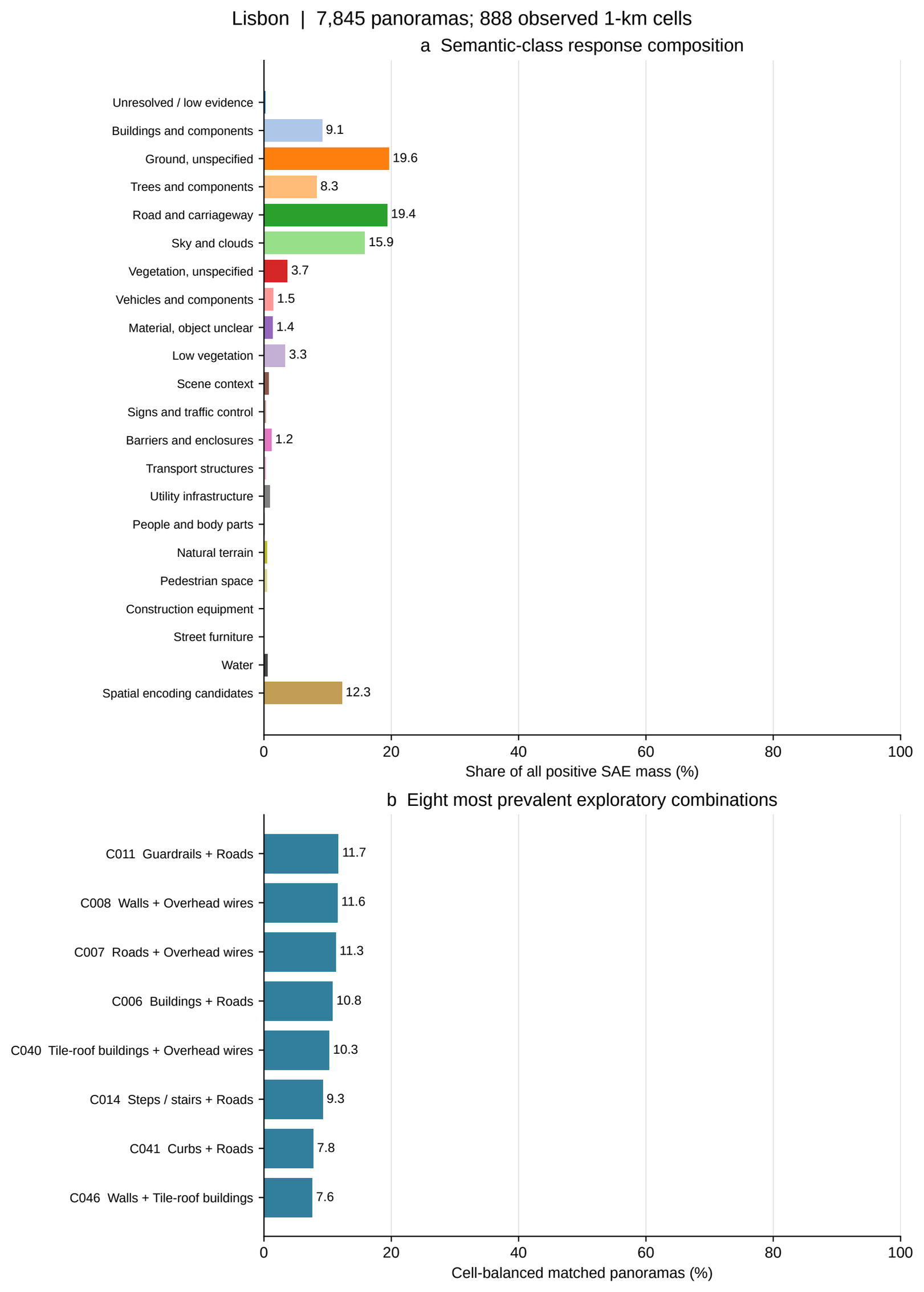

Yogyakarta | 45,043 panoramas; 5,503 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
7.9
Ground, unspecified
17.3
Trees and components
18.6
Road and carriageway
15.0
Sky and clouds
10.9
Vegetation, unspecified
7.8
Vehicles and components
Material, object unclear
1.3
Low vegetation
4.2
Scene context
1.0
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.9
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
17.1
C007 Roads + Overhead wires
9.7
C017 Trees + Overhead wires
9.5
C005 Buildings + Overhead wires
8.8
C016 Sidewalks + Trees
8.6
C010 Shrubs + Trees
7.2
C015 Sidewalks + Overhead wires
5.8
C020 Walls + Trees
5.4
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

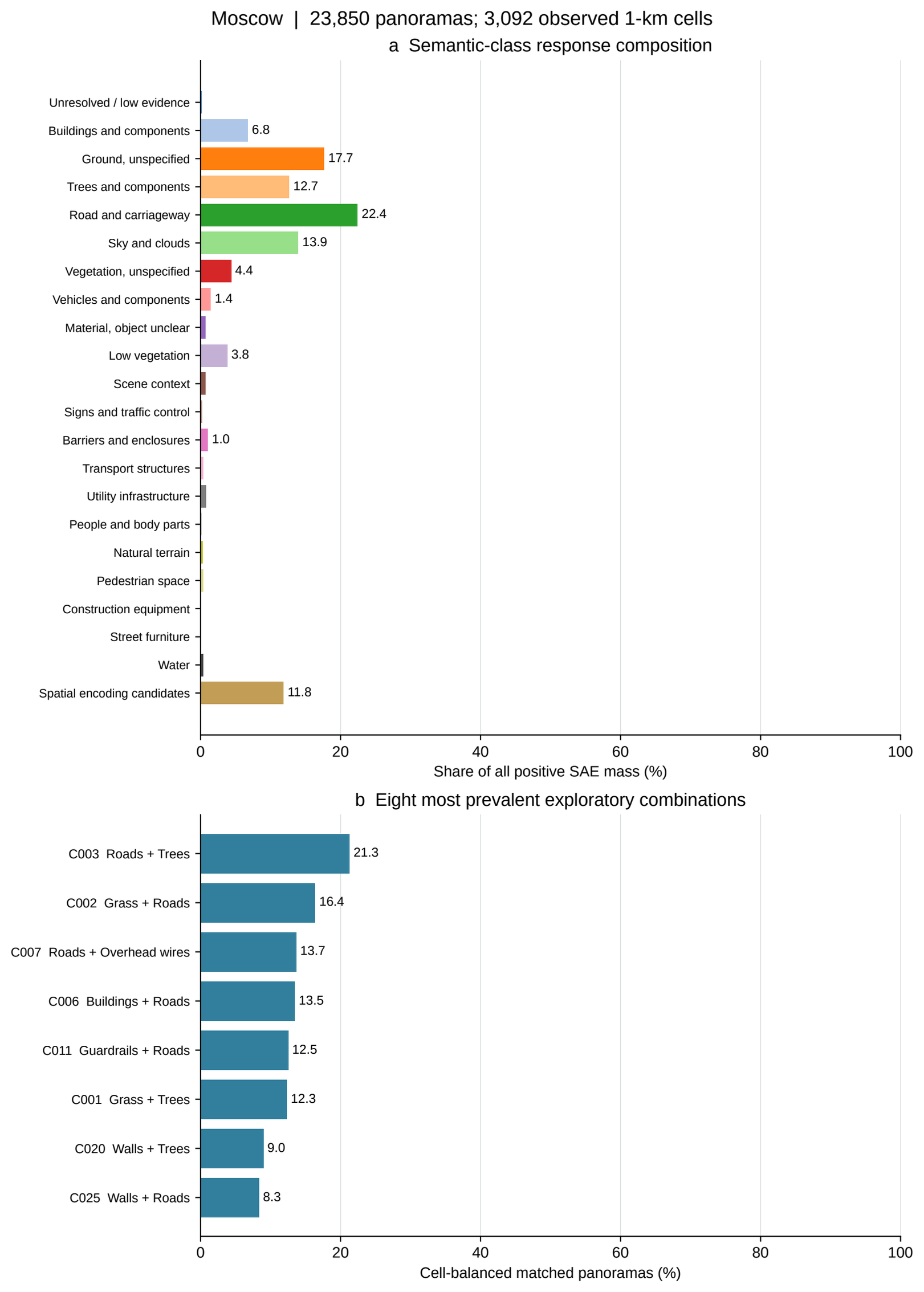

Naples | 12,675 panoramas; 1,419 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
9.9
Ground, unspecified
19.6
Trees and components
9.2
Road and carriageway
17.1
Sky and clouds
14.8
Vegetation, unspecified
4.6
Vehicles and components
1.3
Material, object unclear
1.7
Low vegetation
3.1
Scene context
Signs and traffic control
Barriers and enclosures
1.7
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.2
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C011 Guardrails + Roads
15.0
C024 Buildings + Steps / stairs
12.3
C021 Fences + Buildings
12.3
C057 Fences + Steps / stairs
11.1
C006 Buildings + Roads
8.1
C014 Steps / stairs + Roads
7.7
C009 Walls + Buildings
7.0
C143 Fences + Curbs
6.9
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

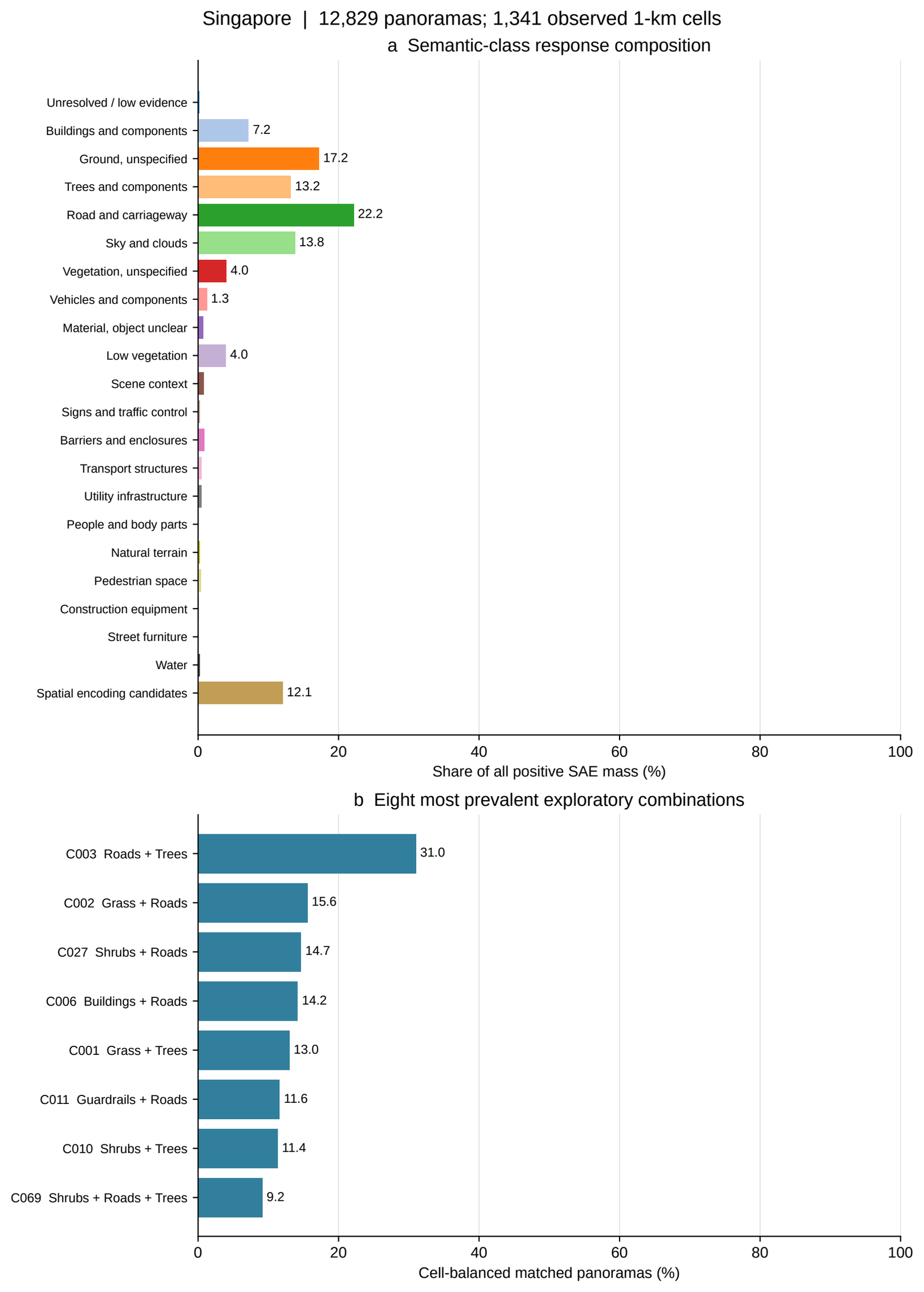

Seoul | 5,914 panoramas; 603 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
14.5
Ground, unspecified
18.6
Trees and components
8.4
Road and carriageway
17.9
Sky and clouds
9.5
Vegetation, unspecified
2.6
Vehicles and components
2.9
Material, object unclear
2.5
Low vegetation
1.7
Scene context
3.2
Signs and traffic control
Barriers and enclosures
1.5
Transport structures
Utility infrastructure
1.5
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.8
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C005 Buildings + Overhead wires
23.6
C023 Buildings + Billboards
18.3
C039 Billboards + Overhead wires
17.5
C006 Buildings + Roads
12.2
C011 Guardrails + Roads
11.7
C086 Buildings + Billboards + Overhead wires
11.6
C030 Roads + Billboards
10.2
C024 Buildings + Steps / stairs
9.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

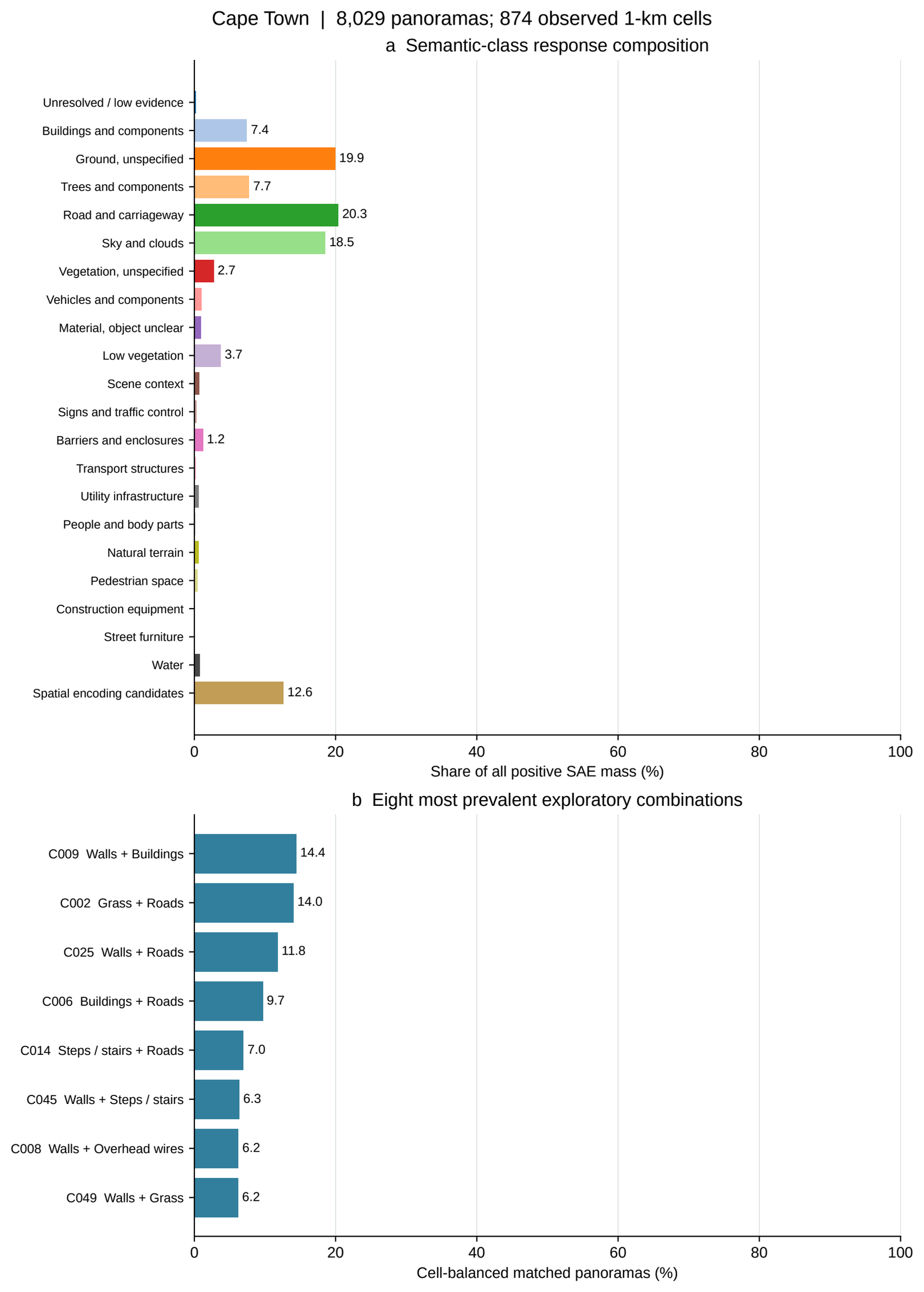

Monterrey | 8,750 panoramas; 954 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
8.1
Ground, unspecified
20.0
Trees and components
7.7
Road and carriageway
20.6
Sky and clouds
16.1
Vegetation, unspecified
2.7
Vehicles and components
2.0
Material, object unclear
1.0
Low vegetation
2.6
Scene context
Signs and traffic control
Barriers and enclosures
1.1
Transport structures
Utility infrastructure
1.8
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.5
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C007 Roads + Overhead wires
28.9
C005 Buildings + Overhead wires
27.0
C008 Walls + Overhead wires
16.9
C015 Sidewalks + Overhead wires
15.5
C047 Roads + Hills / mountains
13.5
C009 Walls + Buildings
12.0
C022 Fences + Overhead wires
11.0
C012 Buildings + Sidewalks
10.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

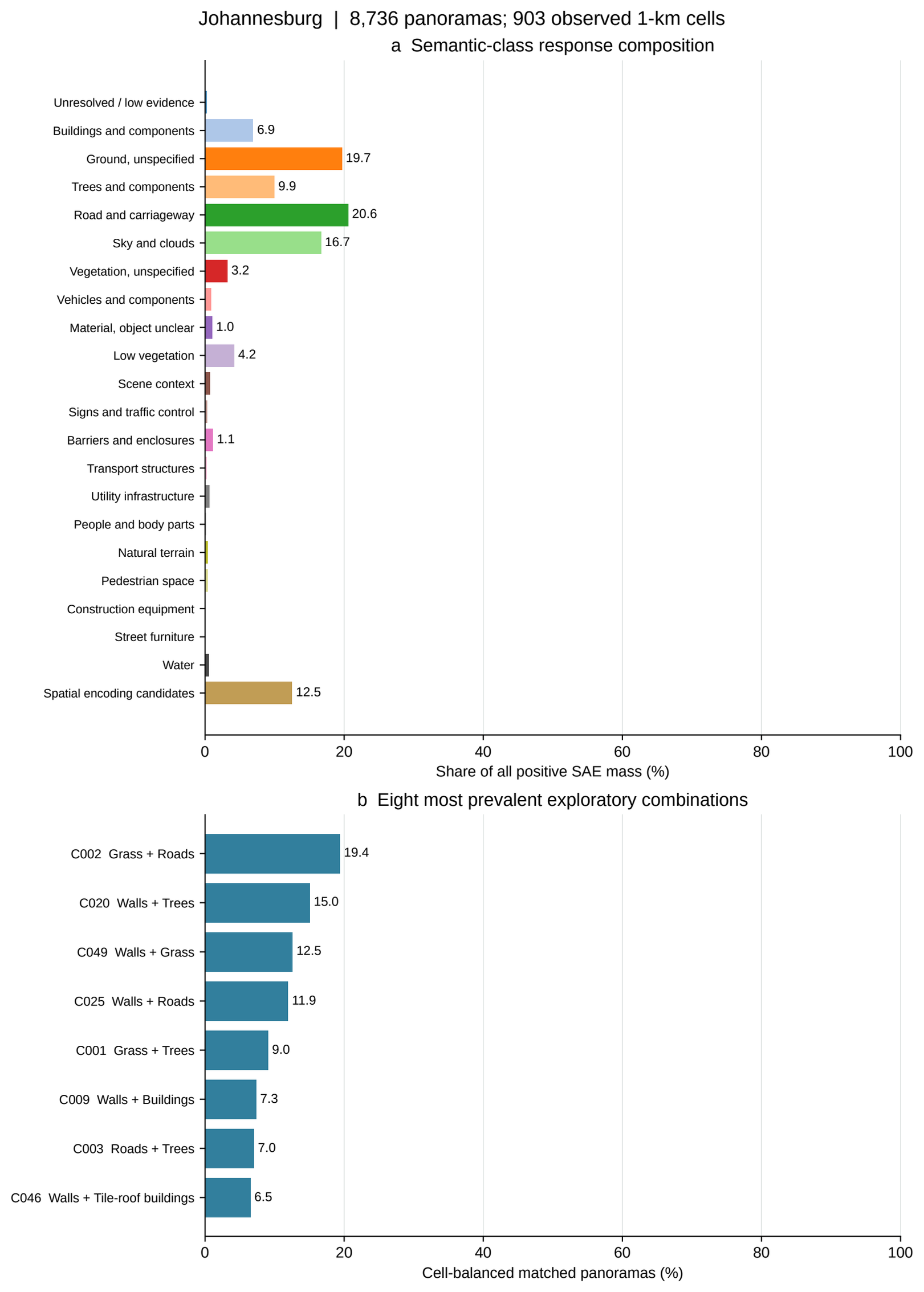

Lagos | 12,221 panoramas; 1,283 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
9.7
Ground, unspecified
21.8
Trees and components
5.9
Road and carriageway
14.9
Sky and clouds
18.2
Vegetation, unspecified
3.2
Vehicles and components
2.0
Material, object unclear
1.6
Low vegetation
2.9
Scene context
Signs and traffic control
Barriers and enclosures
1.4
Transport structures
Utility infrastructure
1.3
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.8
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C008 Walls + Overhead wires
32.7
C005 Buildings + Overhead wires
31.1
C009 Walls + Buildings
29.1
C032 Walls + Buildings + Overhead wires
21.7
C033 Steps / stairs + Overhead wires
16.4
C045 Walls + Steps / stairs
11.6
C024 Buildings + Steps / stairs
11.5
C018 Grass + Overhead wires
10.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

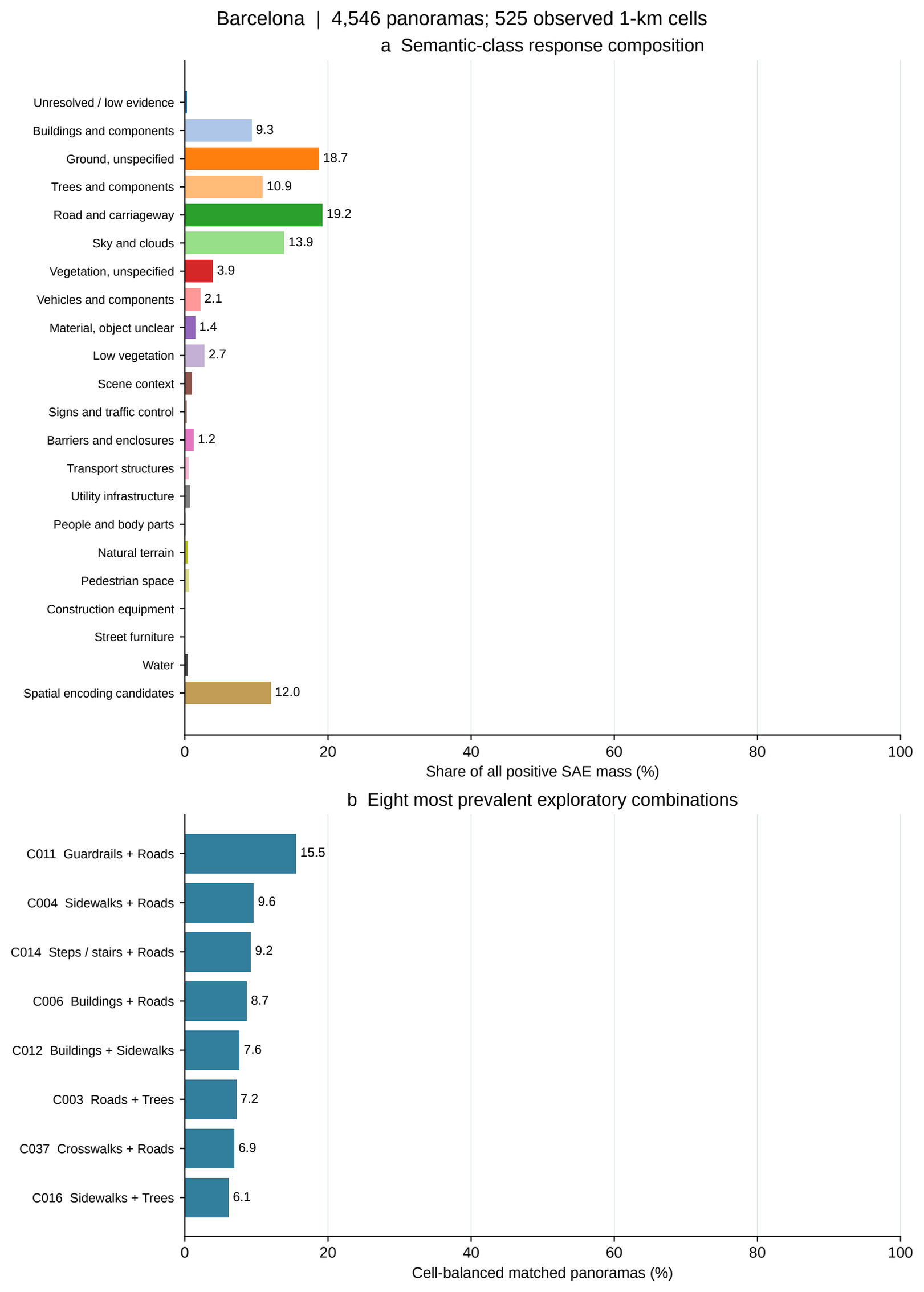

Lisbon | 7,845 panoramas; 888 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
9.1
Ground, unspecified
19.6
Trees and components
8.3
Road and carriageway
19.4
Sky and clouds
15.9
Vegetation, unspecified
3.7
Vehicles and components
1.5
Material, object unclear
1.4
Low vegetation
3.3
Scene context
Signs and traffic control
Barriers and enclosures
1.2
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.3
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C011 Guardrails + Roads
11.7
C008 Walls + Overhead wires
11.6
C007 Roads + Overhead wires
11.3
C006 Buildings + Roads
10.8
C040 Tile-roof buildings + Overhead wires
10.3
C014 Steps / stairs + Roads
9.3
C041 Curbs + Roads
7.8
C046 Walls + Tile-roof buildings
7.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

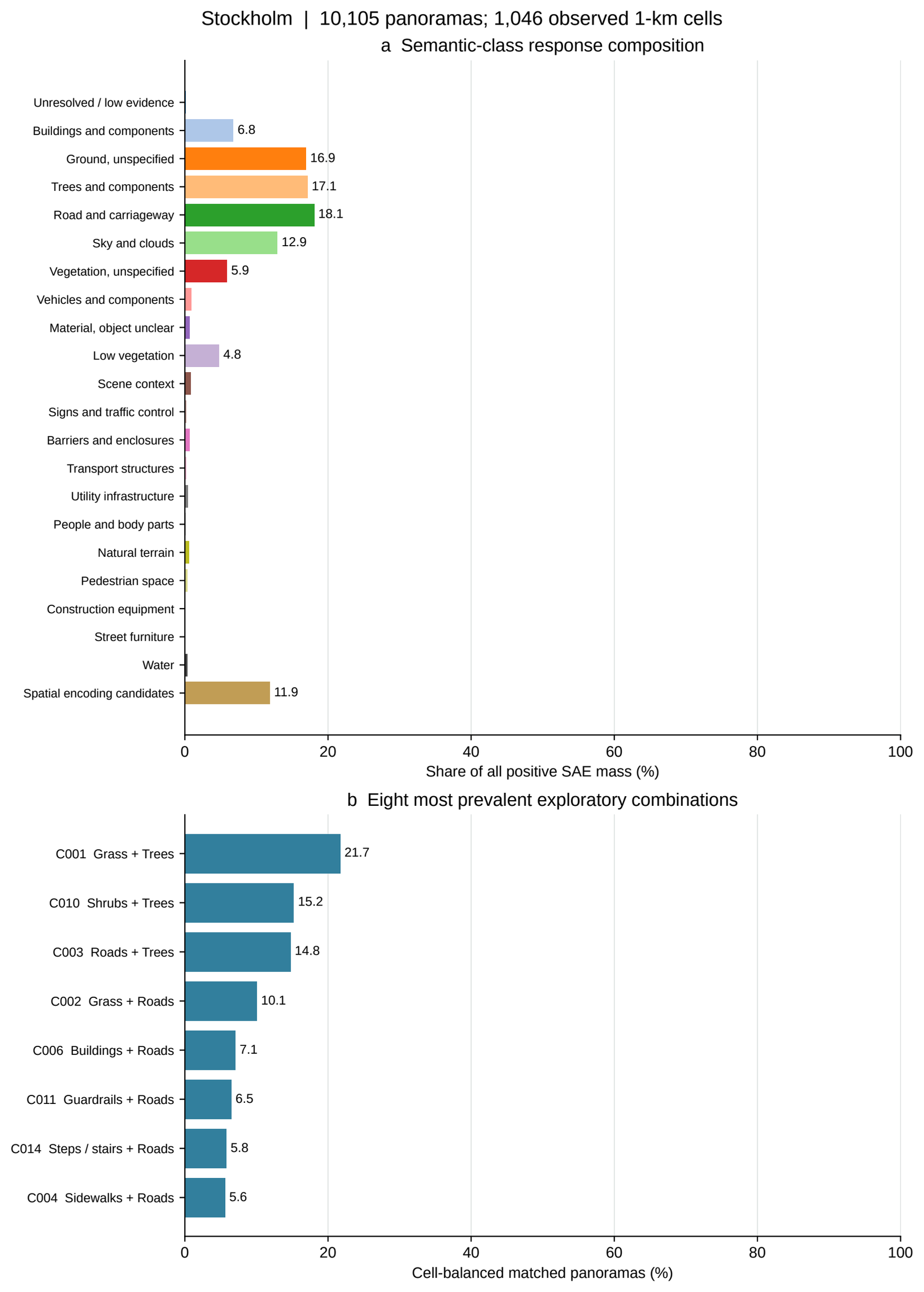

Moscow | 23,850 panoramas; 3,092 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.8
Ground, unspecified
17.7
Trees and components
12.7
Road and carriageway
22.4
Sky and clouds
13.9
Vegetation, unspecified
4.4
Vehicles and components
1.4
Material, object unclear
Low vegetation
3.8
Scene context
Signs and traffic control
Barriers and enclosures
1.0
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.8
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C003 Roads + Trees
21.3
C002 Grass + Roads
16.4
C007 Roads + Overhead wires
13.7
C006 Buildings + Roads
13.5
C011 Guardrails + Roads
12.5
C001 Grass + Trees
12.3
C020 Walls + Trees
9.0
C025 Walls + Roads
8.3
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

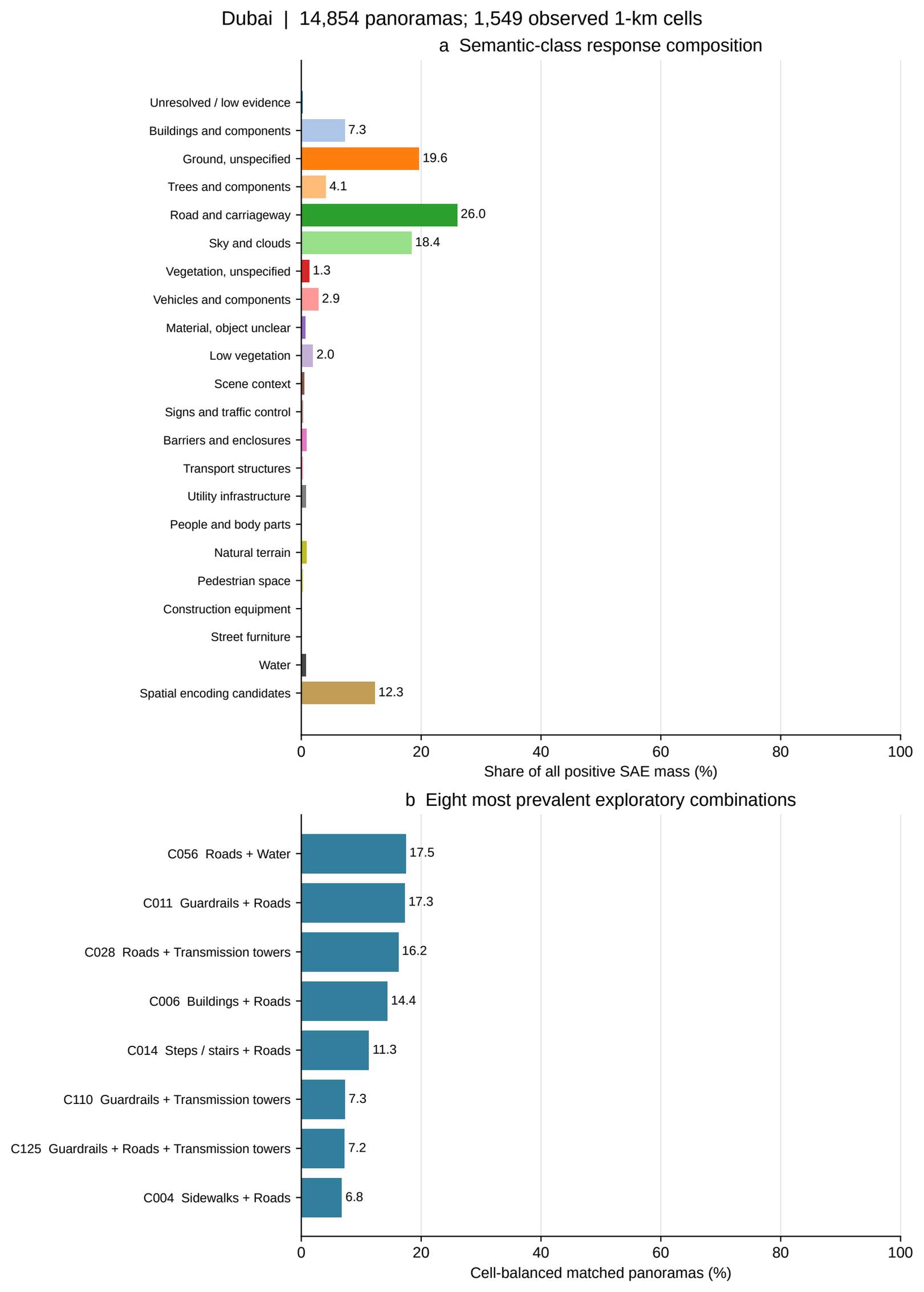

Singapore | 12,829 panoramas; 1,341 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
7.2
Ground, unspecified
17.2
Trees and components
13.2
Road and carriageway
22.2
Sky and clouds
13.8
Vegetation, unspecified
4.0
Vehicles and components
1.3
Material, object unclear
Low vegetation
4.0
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.1
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C003 Roads + Trees
31.0
C002 Grass + Roads
15.6
C027 Shrubs + Roads
14.7
C006 Buildings + Roads
14.2
C001 Grass + Trees
13.0
C011 Guardrails + Roads
11.6
C010 Shrubs + Trees
11.4
C069 Shrubs + Roads + Trees
9.2
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

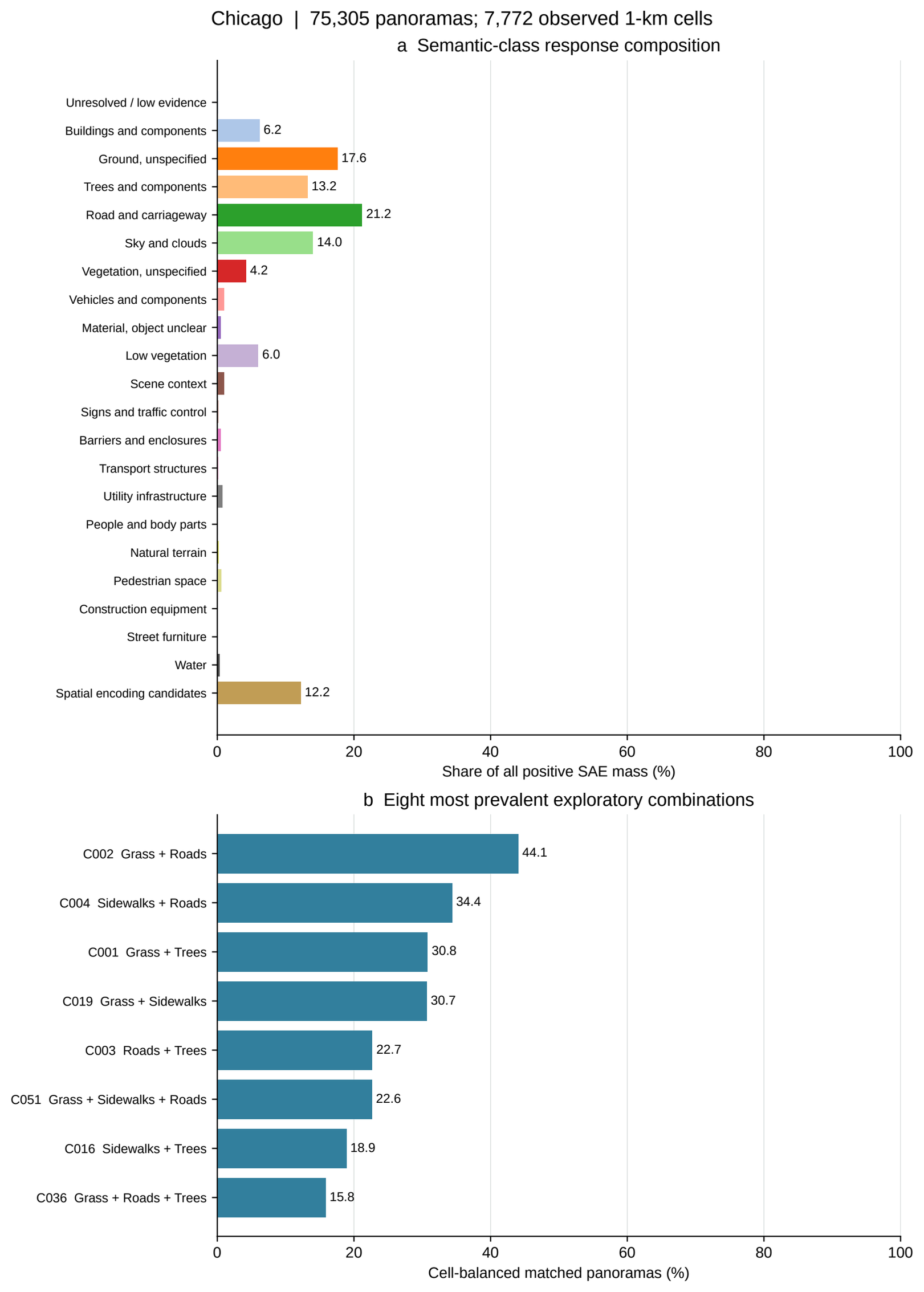

Cape Town | 8,029 panoramas; 874 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
7.4
Ground, unspecified
19.9
Trees and components
7.7
Road and carriageway
20.3
Sky and clouds
18.5
Vegetation, unspecified
2.7
Vehicles and components
Material, object unclear
Low vegetation
3.7
Scene context
Signs and traffic control
Barriers and enclosures
1.2
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.6
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C009 Walls + Buildings
14.4
C002 Grass + Roads
14.0
C025 Walls + Roads
11.8
C006 Buildings + Roads
9.7
C014 Steps / stairs + Roads
7.0
C045 Walls + Steps / stairs
6.3
C008 Walls + Overhead wires
6.2
C049 Walls + Grass
6.2
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

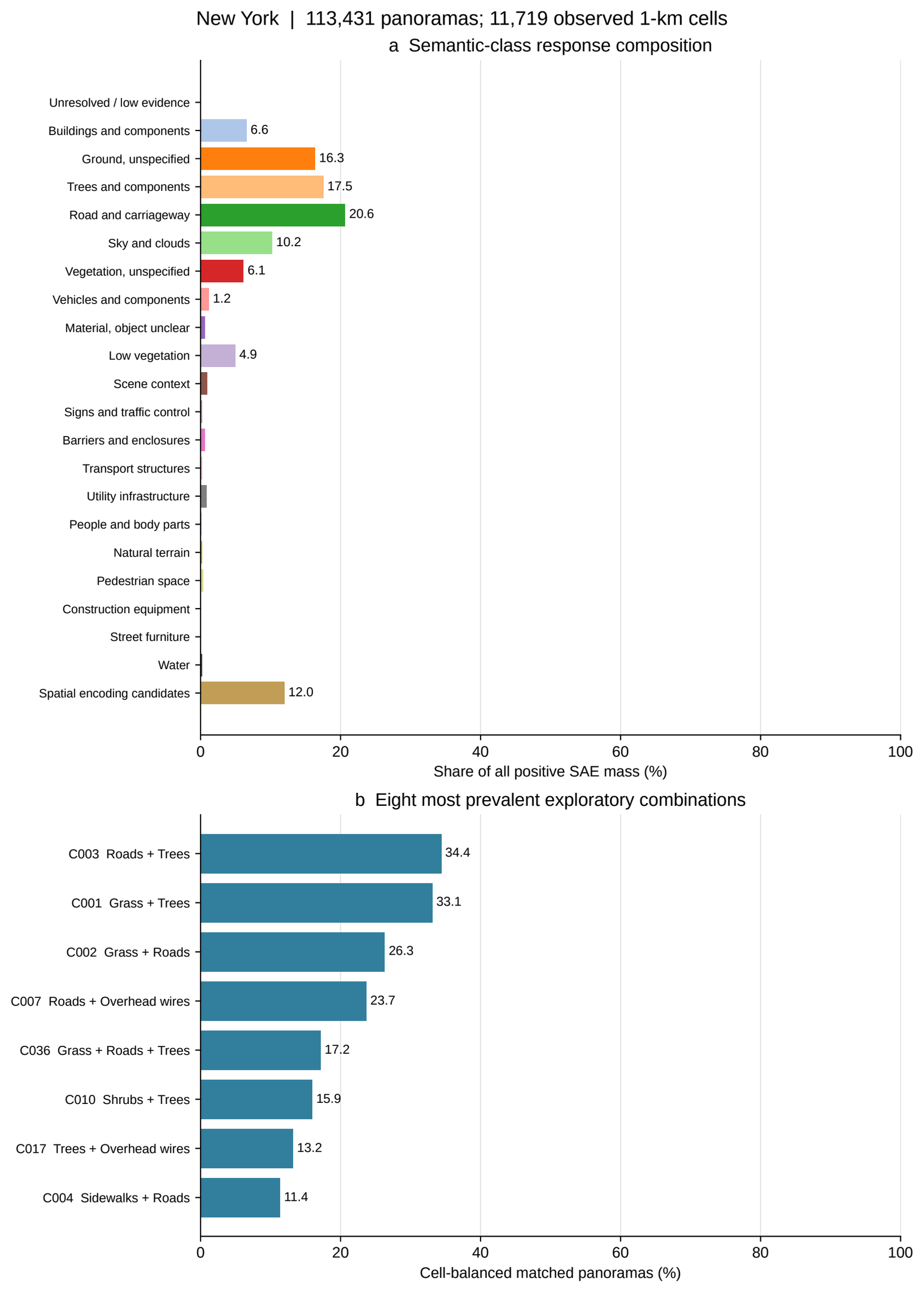

Johannesburg | 8,736 panoramas; 903 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.9
Ground, unspecified
19.7
Trees and components
9.9
Road and carriageway
20.6
Sky and clouds
16.7
Vegetation, unspecified
3.2
Vehicles and components
Material, object unclear
1.0
Low vegetation
4.2
Scene context
Signs and traffic control
Barriers and enclosures
1.1
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.5
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C002 Grass + Roads
19.4
C020 Walls + Trees
15.0
C049 Walls + Grass
12.5
C025 Walls + Roads
11.9
C001 Grass + Trees
9.0
C009 Walls + Buildings
7.3
C003 Roads + Trees
7.0
C046 Walls + Tile-roof buildings
6.5
0
20
40
60
80
100
Cell-balanced matched panoramas (%)

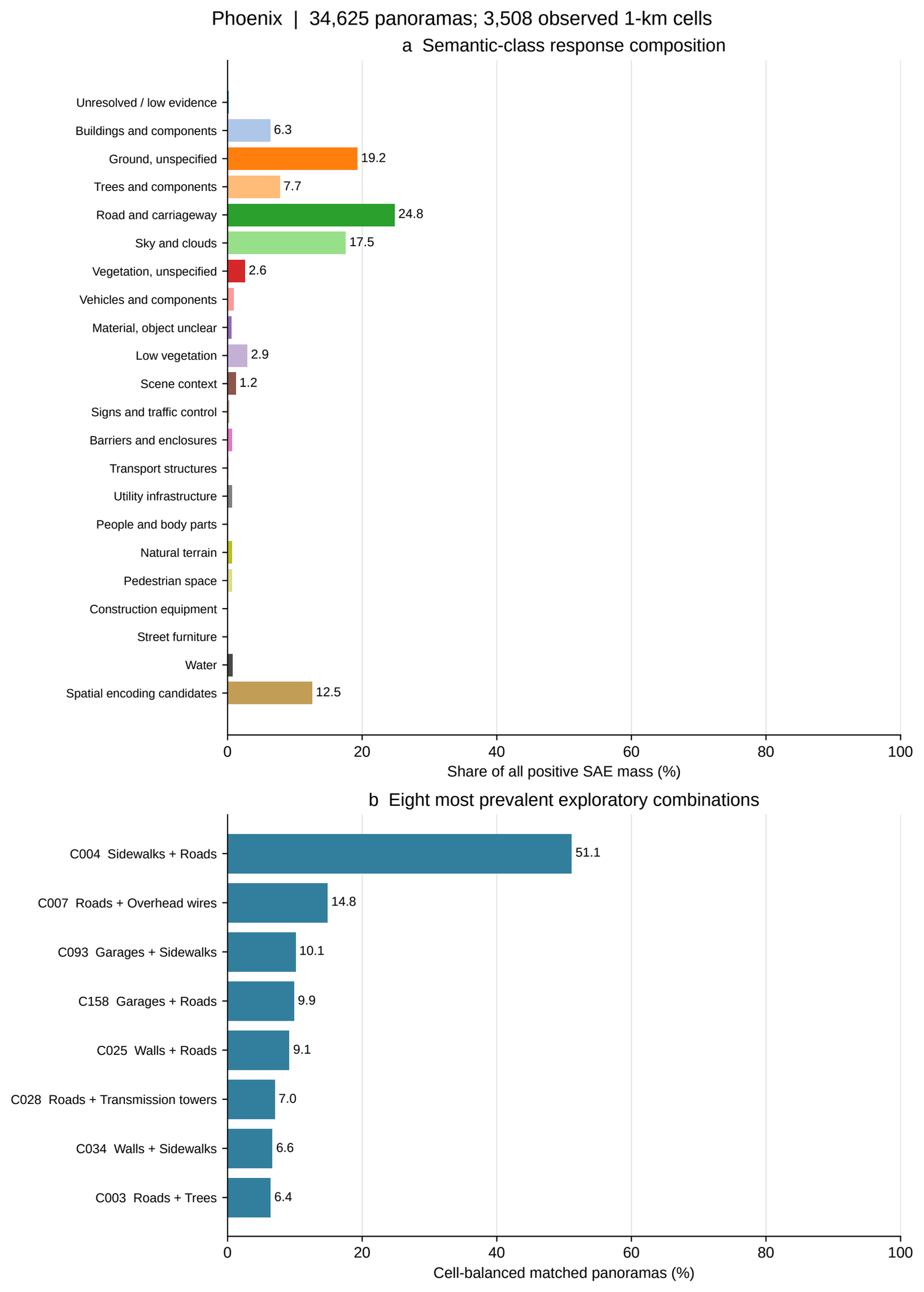

Barcelona | 4,546 panoramas; 525 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
9.3
Ground, unspecified
18.7
Trees and components
10.9
Road and carriageway
19.2
Sky and clouds
13.9
Vegetation, unspecified
3.9
Vehicles and components
2.1
Material, object unclear
1.4
Low vegetation
2.7
Scene context
Signs and traffic control
Barriers and enclosures
1.2
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C011 Guardrails + Roads
15.5
C004 Sidewalks + Roads
9.6
C014 Steps / stairs + Roads
9.2
C006 Buildings + Roads
8.7
C012 Buildings + Sidewalks
7.6
C003 Roads + Trees
7.2
C037 Crosswalks + Roads
6.9
C016 Sidewalks + Trees
6.1
0
20
40
60
80
100
Cell-balanced matched panoramas (%)


Stockholm | 10,105 panoramas; 1,046 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.8
Ground, unspecified
16.9
Trees and components
17.1
Road and carriageway
18.1
Sky and clouds
12.9
Vegetation, unspecified
5.9
Vehicles and components
Material, object unclear
Low vegetation
4.8
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
11.9
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C001 Grass + Trees
21.7
C010 Shrubs + Trees
15.2
C003 Roads + Trees
14.8
C002 Grass + Roads
10.1
C006 Buildings + Roads
7.1
C011 Guardrails + Roads
6.5
C014 Steps / stairs + Roads
5.8
C004 Sidewalks + Roads
5.6
0
20
40
60
80
100
Cell-balanced matched panoramas (%)


Dubai | 14,854 panoramas; 1,549 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
7.3
Ground, unspecified
19.6
Trees and components
4.1
Road and carriageway
26.0
Sky and clouds
18.4
Vegetation, unspecified
1.3
Vehicles and components
2.9
Material, object unclear
Low vegetation
2.0
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.3
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C056 Roads + Water
17.5
C011 Guardrails + Roads
17.3
C028 Roads + Transmission towers
16.2
C006 Buildings + Roads
14.4
C014 Steps / stairs + Roads
11.3
C110 Guardrails + Transmission towers
7.3
C125 Guardrails + Roads + Transmission towers
7.2
C004 Sidewalks + Roads
6.8
0
20
40
60
80
100
Cell-balanced matched panoramas (%)


Chicago | 75,305 panoramas; 7,772 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
Ground, unspecified
Trees and components
Road and carriageway
Sky and clouds
Vegetation, unspecified
Vehicles and components
Material, object unclear
Low vegetation
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
6.2
17.6
13.2
21.2
14.0
4.2
6.0
12.2
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C002 Grass + Roads
C004 Sidewalks + Roads
C001 Grass + Trees
C019 Grass + Sidewalks
C003 Roads + Trees
C051 Grass + Sidewalks + Roads
C016 Sidewalks + Trees
C036 Grass + Roads + Trees
44.1
34.4
30.8
30.7
22.7
22.6
18.9
15.8
0
20
40
60
80
100
Cell-balanced matched panoramas (%)


New York | 113,431 panoramas; 11,719 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.6
Ground, unspecified
16.3
Trees and components
17.5
Road and carriageway
20.6
Sky and clouds
10.2
Vegetation, unspecified
6.1
Vehicles and components
1.2
Material, object unclear
Low vegetation
4.9
Scene context
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.0
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C003 Roads + Trees
34.4
C001 Grass + Trees
33.1
C002 Grass + Roads
26.3
C007 Roads + Overhead wires
23.7
C036 Grass + Roads + Trees
17.2
C010 Shrubs + Trees
15.9
C017 Trees + Overhead wires
13.2
C004 Sidewalks + Roads
11.4
0
20
40
60
80
100
Cell-balanced matched panoramas (%)


Phoenix | 34,625 panoramas; 3,508 observed 1-km cells
a Semantic-class response composition
Unresolved / low evidence
Buildings and components
6.3
Ground, unspecified
19.2
Trees and components
7.7
Road and carriageway
24.8
Sky and clouds
17.5
Vegetation, unspecified
2.6
Vehicles and components
Material, object unclear
Low vegetation
2.9
Scene context
1.2
Signs and traffic control
Barriers and enclosures
Transport structures
Utility infrastructure
People and body parts
Natural terrain
Pedestrian space
Construction equipment
Street furniture
Water
Spatial encoding candidates
12.5
0
20
40
60
80
100
Share of all positive SAE mass (%)
b Eight most prevalent exploratory combinations
C004 Sidewalks + Roads
51.1
C007 Roads + Overhead wires
14.8
C093 Garages + Sidewalks
10.1
C158 Garages + Roads
9.9
C025 Walls + Roads
9.1
C028 Roads + Transmission towers
7.0
C034 Walls + Sidewalks
6.6
C003 Roads + Trees
6.4
0
20
40
60
80
100
Cell-balanced matched panoramas (%)